

Black Body Radiation and Climate Forcing

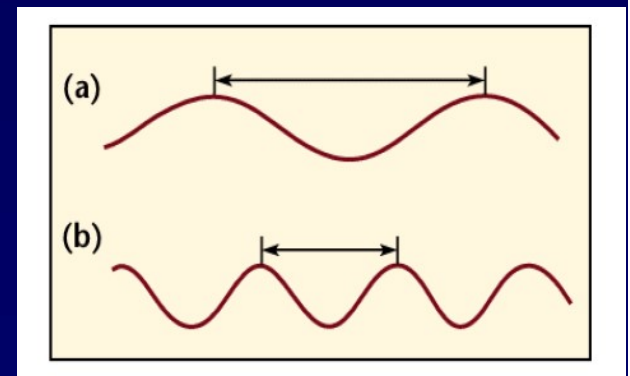


Prof. A.K. Chaubey
akchaubey@iitism.ac.in

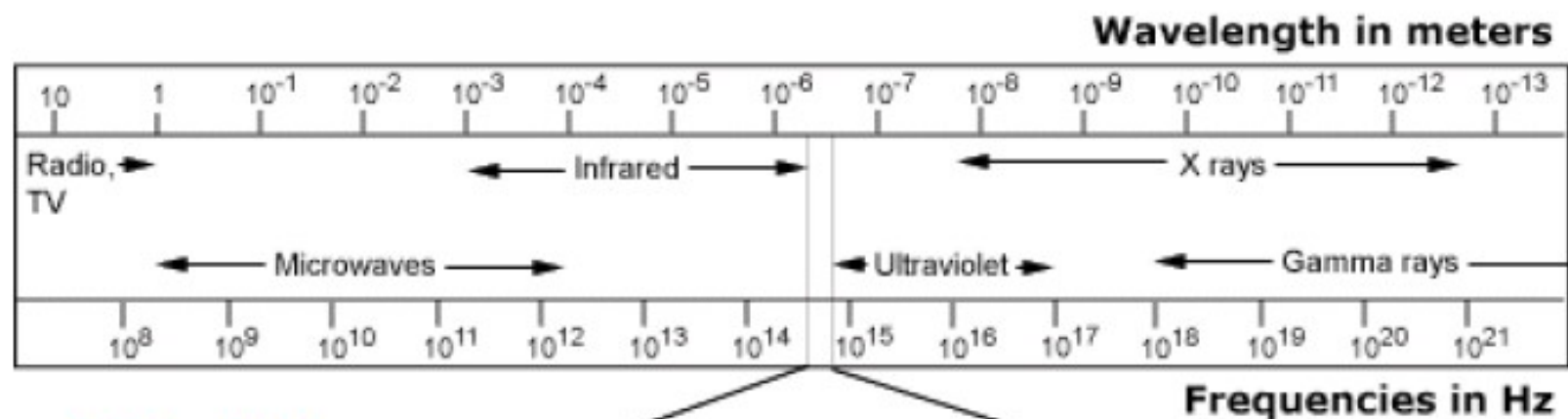
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L22: 04 Mar. 2024
L23: 06 Mar. 2024
L24: 11 Mar. 2024
L25: 13 Mar. 2024
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Radiation

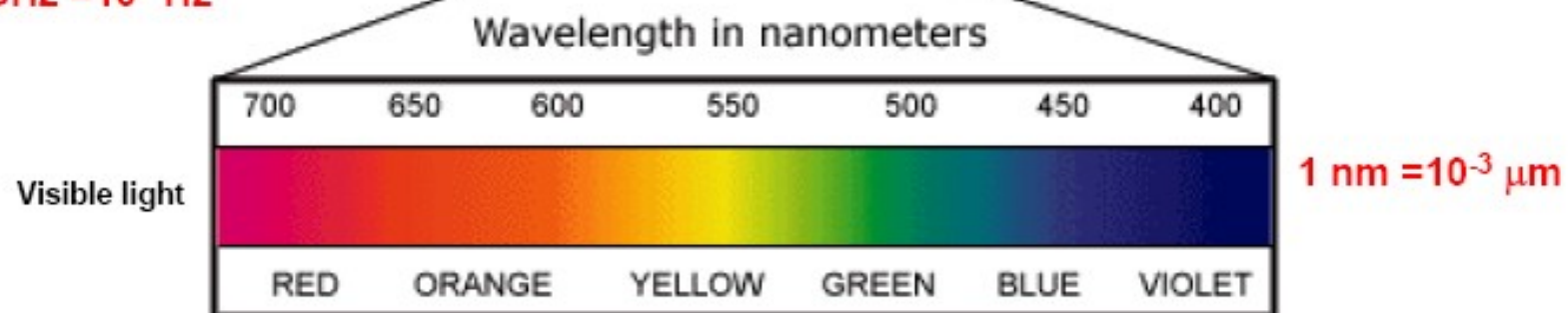
- An object at temperature T can emit or absorb light waves of several frequencies. The distribution of the energy exchanged by the object with its external environment depends on the temperature T .
- When two objects at different temperatures are in contact, thermal energy (heat) is transferred from the hot object to the cold object.
- By contrast, radiation is energy that is carried by an electromagnetic wave. In this case, energy is transferred by emission and absorption of light waves. Hence, by definition, thermal radiation is the electromagnetic radiation emitted by any object at non-zero temperature T .
- Electromagnetic waves move through space at a constant speed $c = \sim 3 \times 10^8 \text{ m s}^{-1}$
- Sunlight, microwaves, heat from a fire, radio waves, ultraviolet rays, x rays, gamma rays.
- The different types of radiation are distinguished by their wavelength λ .
 - a = long wavelength → less Energy
 - b = short wavelength → more Energy



The various forms of radiation are organized according to their wavelengths (and hence energy levels), creating the **electromagnetic spectrum**.



1GHz = 10^9 Hz



1 nm = 10^{-3} μ m

Origin of Radiation

- In 1900, Max Planck laid the foundations of quantum physics by studying the black body emission spectrum within the theory of quanta.
- Max Karl Ernst Ludwig Planck, was a German physicist. He founded quantum physics in 1900 with his fundamental hypothesis on the theory of quanta.
- He formulated the fundamental hypothesis according to which the energy generated by a periodic movement of frequency ν (rotation or vibration) has a discontinuous structure.
- Consequently, radiant energy can only exist as bundles or quanta of energy $h\nu$. The number h is a universal constant known as the Planck constant.

In 1905, **Albert Einstein** stated that light is made of particles subsequently called photons, each of which has an energy $h\nu$. Radiation results from *electronic transitions* between discrete levels of atomic or molecular systems. The energy exchanged during these transitions corresponds to *photon absorption and emission processes*. The energy E , angular frequency ω , frequency ν and wavelength λ of the photon are related by the following relations:

$$\begin{cases} E = h\nu = \hbar\omega \\ E = \frac{hc}{\lambda} \end{cases} \Rightarrow \begin{cases} \nu = \frac{\omega}{2\pi} = \frac{c}{\lambda} \\ \hbar = \frac{h}{2\pi} \end{cases}$$

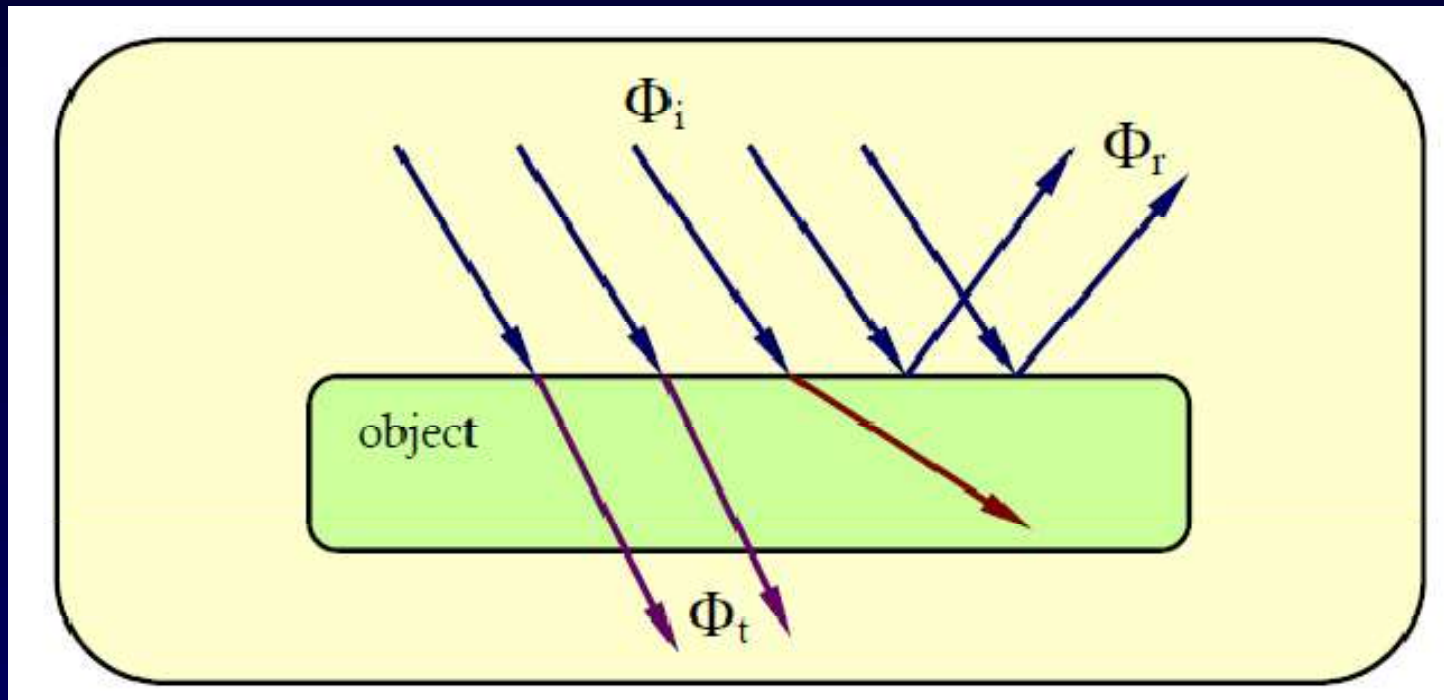
Where, E is expressed in joules (J), ν in hertz (Hz), ω in radian per second and λ in meters (m). The quantity c designates the *speed of light* in a vacuum and $\hbar$ is the *h-bar (or reduced) Planck constant*.

Numerically, $h = 6.62606896 \cdot 10^{-34}$ J·s; h-bar ($\hbar$) = $1.054571628 \cdot 10^{-34}$ J·s.
 $c = 299792458$ m·s⁻¹ (exact value).

At the microscopic scale, it is convenient to use the *electron-volt* (eV) as a unit of energy, where:

$$1 \text{ eV} = 1.602179487 \cdot 10^{-19} \text{ J.}$$

Decomposition of an incident radiant flux Φ_i



- By definition, *radiant flux*, denoted by Φ , is the power emitted by a source throughout the space in which it can radiate. Radiant flux is expressed in Watts (W).
- Let us consider an object receiving an incident energy flux Φ_i . The surface of the object is chosen to allow radiation reflection, absorption and transmission.

Reflection, Transmission and Absorption of EMR

According to the law of conservation of energy, we can write:

$$\Phi_i = \Phi_r + \Phi_a + \Phi_t \dots\dots\dots(1)$$

In this relation, Φ_r , Φ_a and Φ_t are the *reflected radiant flux*, *absorbed radiant flux* and *transmitted radiant flux*, respectively. Let us consider ρ , α and ζ as the *coefficients of reflection*, *absorption* and *transmission*, respectively, of the radiant flux. These quantities by definition can be expressed as:

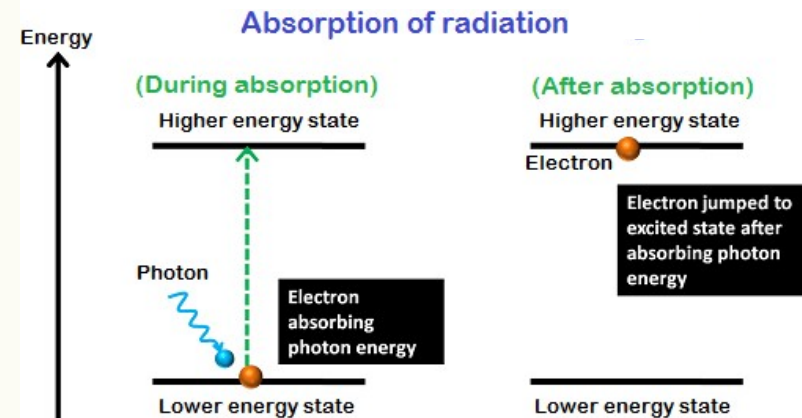
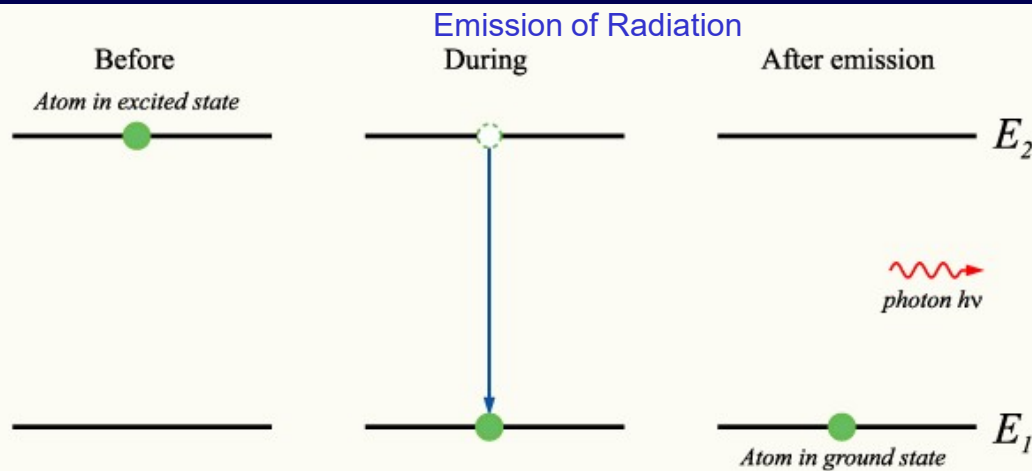
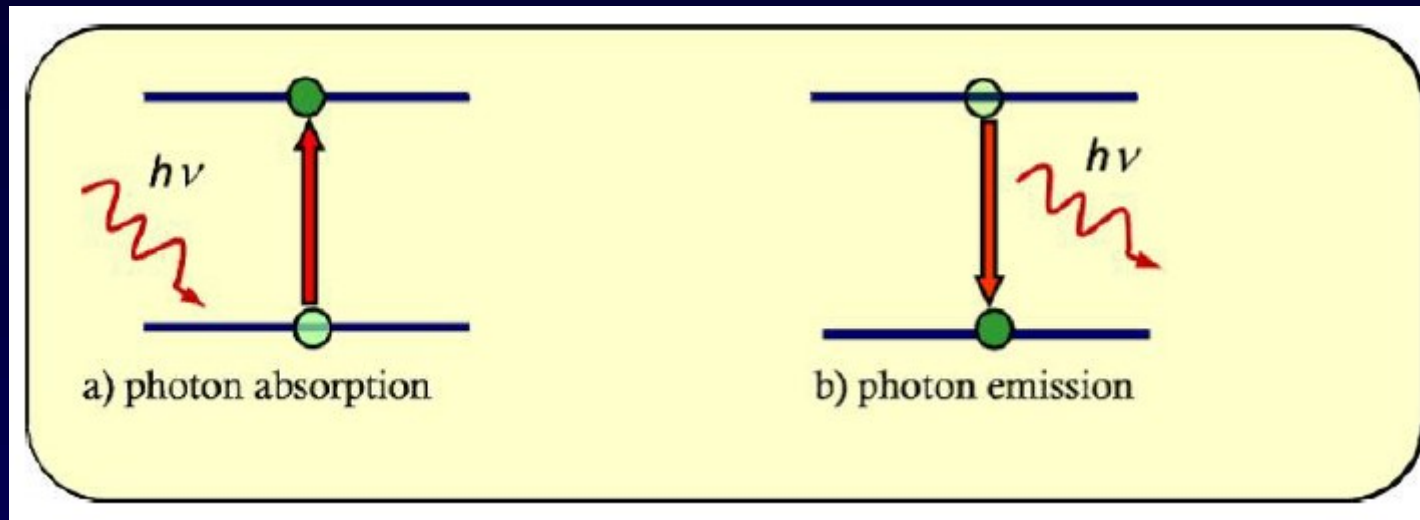
$$\rho = \frac{\Phi_r}{\Phi_i} \quad ; \quad \alpha = \frac{\Phi_a}{\Phi_i} \quad ; \quad \tau = \frac{\Phi_t}{\Phi_i} \dots\dots\dots(2)$$

By dividing equation (1) by Φ_i and using equation (2), we get:

$$\rho + \alpha + \tau = 1$$

Coefficients ρ , α and ζ characterize the behavior of an object that is subjected to radiation. It is worth noting that the absorption coefficient α is the most important parameter. This coefficient measures the proportion in which incident electromagnetic radiation is converted into thermal energy.

Photon absorption and emission processes



Classification of objects

Objects susceptible to exchange energy are classified into three categories:

- Transparent objects that allow radiation to pass through without attenuation. This is the case with glass, transparent plastic material, etc.
- Opaque objects that absorb radiation and get heated. This is the case with solid bodies (metals, rocks, etc.), cardboard and some viscous liquids, such as paint.
- Translucent objects that absorb a part of the radiation and allow the rest to pass through. For these objects, radiation propagation is accompanied by absorption that increases the energy of the medium. A familiar example is that of oil.

Black Body and Gray Body

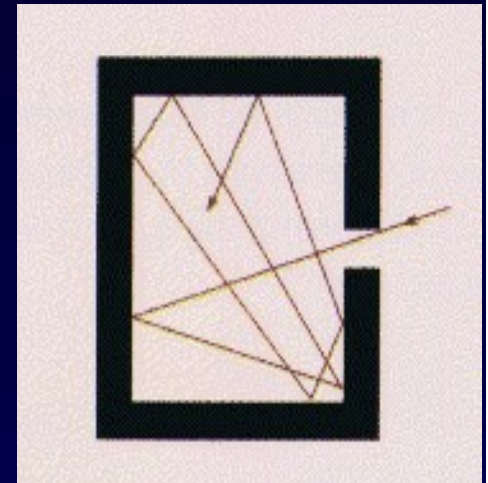
A **black body** is an ideal object that has the specific property of perfectly absorbing the electromagnetic radiations irrespective of their frequency.

The adjective “black” highlights only the fact that the object absorbs all the radiations of the visible spectrum so that it appears to be black.

In other words, a black body is an ideal body which allows the whole of the incident radiation to pass into itself (without reflecting the energy) and absorbs within itself the whole incident radiation. This property is valid for radiation corresponding to all wavelengths and to all angles of incidence. Therefore, the black body is an ideal absorber of incident radiation.

Any object whose absorption coefficient $\alpha = 1$, is termed as black body.

Gray Body: Please note that a gray body is not necessarily gray. This term designates any object whose absorption coefficient is $\alpha < 1$.



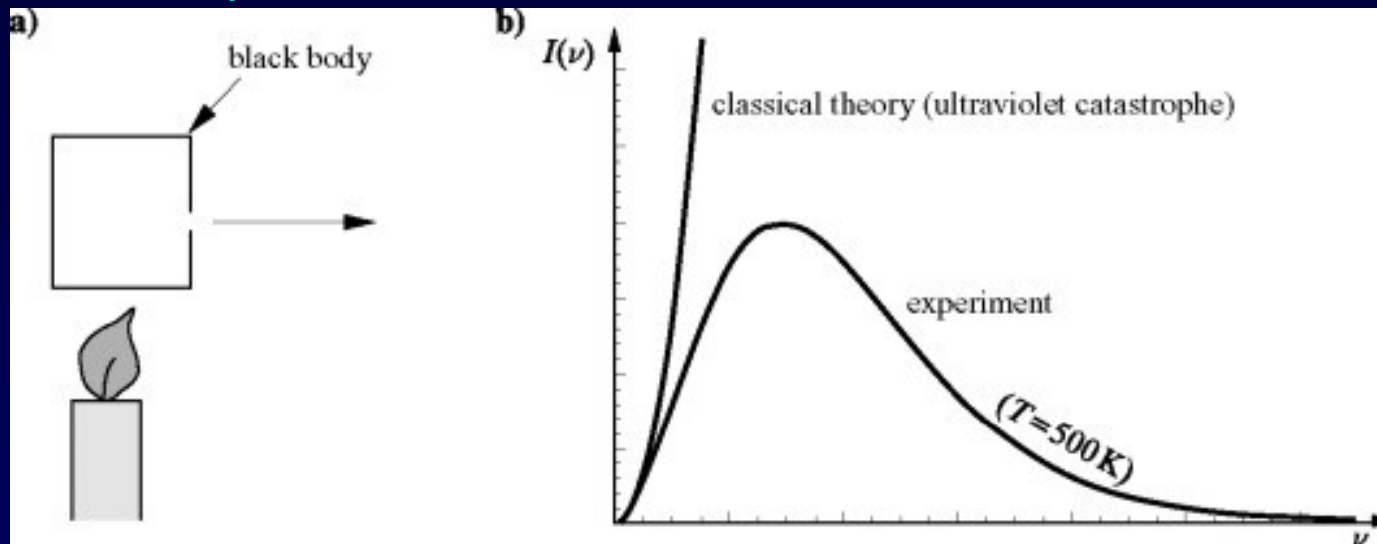
Black body and Gray body

There are two types of bodies:

- Gray bodies for which $\alpha < 1$
- Black bodies for which $\alpha = 1$.

The Magic of Quantum Mechanics

Max Planck wanted to understand black body radiation. He used a black body of a box shape with a small hole in it.



He heated the box up, waited for the system to reach a stationary state (at a fixed temperature) and see what kind of electromagnetic radiation (intensity as a function of frequency) comes out of the hole. He calculated correctly that the black body would emit electromagnetic radiation having a distribution of frequencies. However, the larger the frequency the larger its intensity, leading to what is known as ultraviolet catastrophe. There is a serious discrepancy between the results of classical theory and the experiment, especially for large frequencies. Only after assuming the existence of energy quanta, theory and experiment can be reconciled.

Black-Body Emission Spectrum

What is Radiant Exitance or Radiant Emittance?

- Radiant exitance or Radiant Emittance is the radiant flux emitted by a surface per unit area,
- The SI unit of radiant exitance is the watt per square metre (W/m^2), while that of spectral exitance in frequency is the watt per square metre per hertz ($\text{W}\cdot\text{m}^{-2}\cdot\text{Hz}^{-1}$) and that of spectral exitance in wavelength is the watt per square metre per metre ($\text{W}\cdot\text{m}^{-3}$). Radiant exitance is often called "intensity" in branches of physics other than radiometry, but in radiometry this usage leads to confusion with radiant intensity.
- Radiant exitance of a *surface*, denoted M_e ("e" for "energetic") is defined as:

$$M_e = \frac{\partial \Phi_e}{\partial A},$$

Where, ∂ is the partial derivative symbol; Φ_e is the radiant flux emitted; A is the area.

Radiant Exitance or Radiant Emittance.....Contd.

Radiant exitance of a black body surface (M_e^o), according to the Stefan-Boltzmann Law, is given by:

$$M_e^o = \sigma T^4$$

Where, σ is Stefan - Boltzmann constant, and T is the temperature of the black body surface.

For a gray body surface, the radiant exitance (M_e) is given by:

$$M_e = \epsilon M_e^o = \epsilon \sigma T^4$$

Where, ϵ is the emissivity of the surface.

Please note that Radiant Exitance of a black body *surface* is denoted as M_e^o ("e" for "energetic" and "o" for black body).

Spectral Exitance / Spectral Emittance

Spectral exitance / spectral emittance is the radiant exitance of a surface per unit frequency or wavelength, depending on whether the spectrum is taken as a function of frequency or of wavelength. In frequency of Spectral exitance of a *surface*, denoted $M_{e,\nu}$, in frequency is given as:

$$M_{e,\nu} = \frac{\partial M_e}{\partial \nu}$$

Spectral exitance / spectral emittance of a *surface*, denoted $M_{e,\lambda}$, is given in wavelength as:

$$M_{e,\lambda} = \frac{\partial M_e}{\partial \lambda}$$

Where, ν is the frequency, and λ is the wavelength.

Spectral Exitance / Spectral Emittance.....Contd.

- The Spectral Exitance or Spectral Emittance is the radiant exitance of a surface per unit frequency or wavelength, depending on whether the spectrum is taken as a function of frequency or of wavelength.
- Spectral Exitance / Spectral Emittance is computed using formulae of Planck's law.
- It also provides formula for the computation of the spectral exitance for a real surface (Gray-body surface).

Planck's Radiation Law: Spectral Distribution of Emissive Power

Planck's law describes the electromagnetic radiation emitted by a black body in thermal equilibrium at a definite temperature.

The law is named after Max Planck, who originally proposed it in 1900. Planck's blackbody formulation to express the spectral exitance of the black body around a given frequency ($M_{e,\nu}^{\circ}$) or wavelength ($M_{e,\lambda}^{\circ}$) are given as :

$$M_{e,\nu}^{\circ} = \pi L_{e,\Omega,\nu}^{\circ} = \frac{2\pi h \nu^3}{c^2} \frac{1}{e^{\frac{h\nu}{kT}} - 1},$$
$$M_{e,\lambda}^{\circ} = \pi L_{e,\Omega,\lambda}^{\circ} = \frac{2\pi h c^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda kT}} - 1},$$

Where, h is the Planck constant ($h = 6.62606896 \cdot 10^{-34}$ J.s), ν is the frequency; λ is the wavelength; k is the Boltzmann constant; c is the speed of light in the medium; T is the temperature of that surface, λ is the wavelength, L° is the radiance of the black body

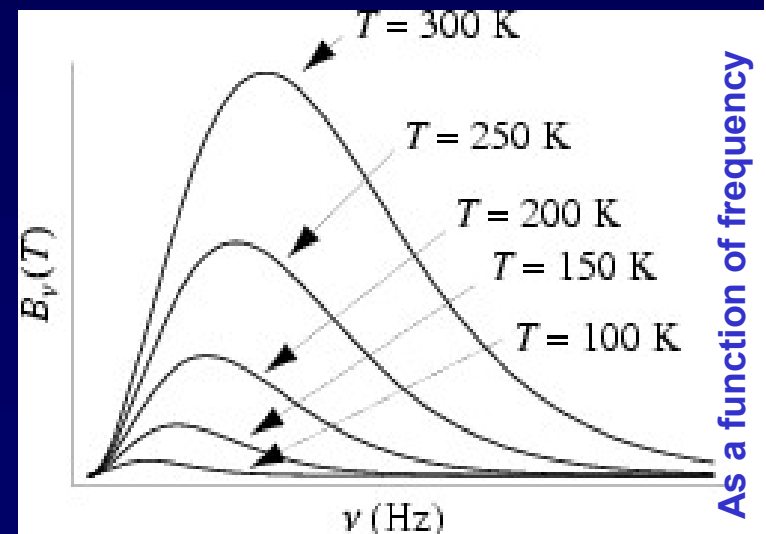
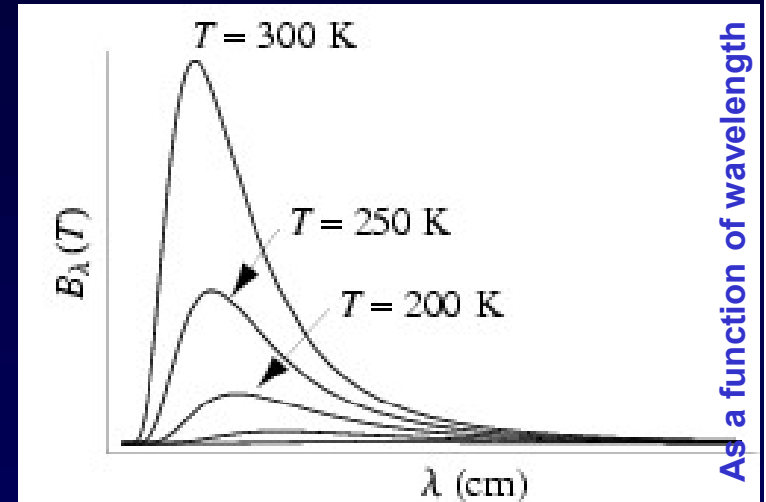
Planck's Law: Spectral Distribution of Emissive Power

Planck's formulation to express the spectral exitance of the gray-body around a given frequency ($M_{e,\nu}$) or wavelength ($M_{e,\lambda}$) are given as :

$$M_{e,\nu} = \varepsilon M_{e,\nu}^{\circ} = \frac{2\pi h \varepsilon \nu^3}{c^2} \frac{1}{e^{\frac{h\nu}{kT}} - 1},$$

$$M_{e,\lambda} = \varepsilon M_{e,\lambda}^{\circ} = \frac{2\pi h \varepsilon c^2}{\lambda^5} \frac{1}{e^{\frac{hc}{\lambda kT}} - 1}.$$

Where, ε is the emissivity of the surface.



Isotherms of a black body: experimental Facts

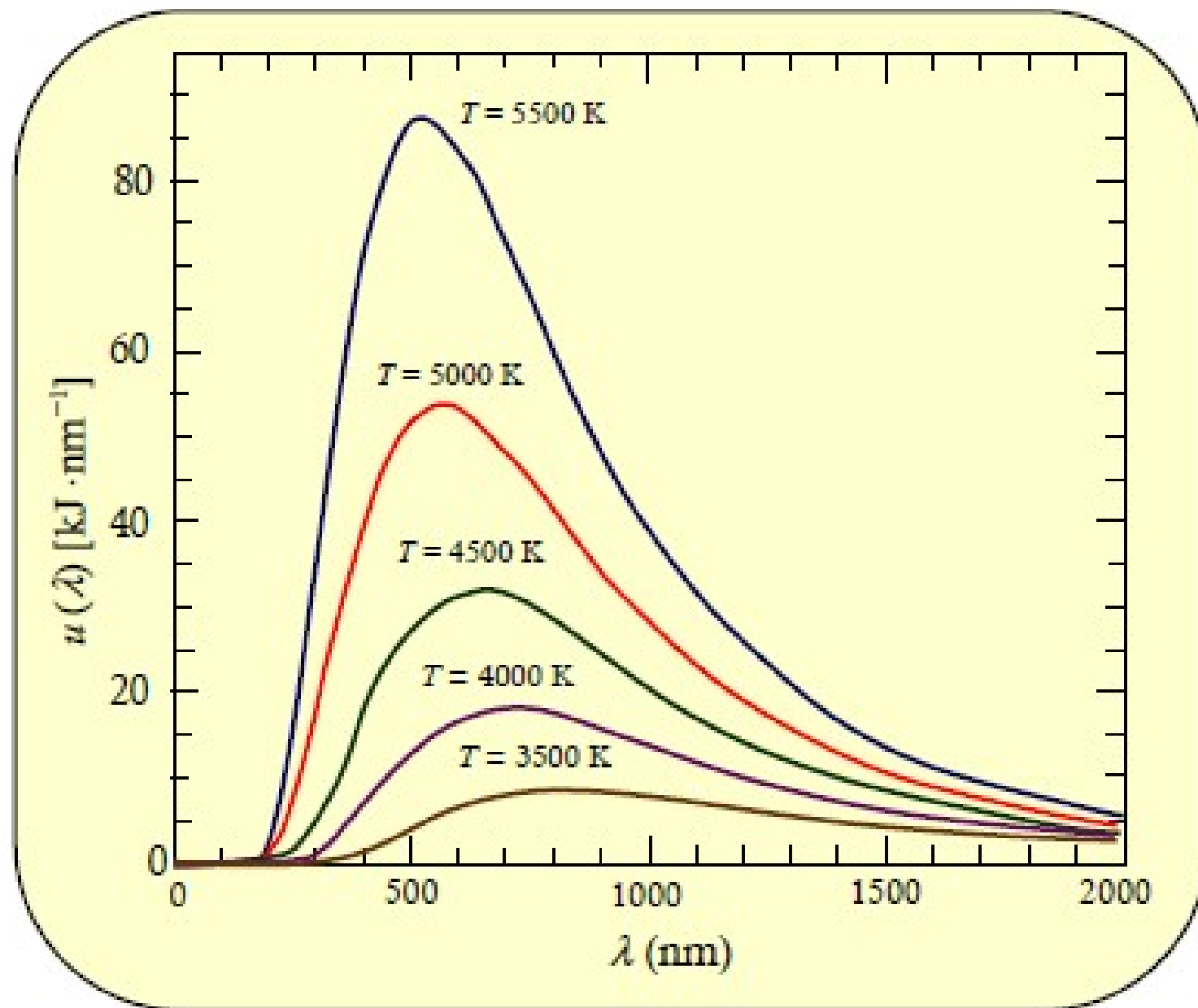
By definition, the *electromagnetic energy density* denoted by du in the band of angular frequency between ω and $\omega+d\omega$ (or of wavelength between λ and $\lambda+d\lambda$) is given by the expression:

$$du = u(\omega) d\omega = u(\lambda) d\lambda \text{ -----(1)}$$

The physical quantity $u(\omega)$ or $u(\lambda)$ is called the *spectral density of electromagnetic energy*. $u(\omega)$ is expressed in $\text{J} \cdot \text{rad}^{-1} \cdot \text{s}$ and $u(\lambda)$ in $\text{J} \cdot \text{m}^{-1}$.

Let us study the variation of the spectral density of electromagnetic energy depending on wavelength λ for each temperature T of the black body. Experience shows that these are asymmetrical curves known as *black body isotherms*. For each temperature value T , there is a corresponding curve that reaches a maximum for a specific wavelength value denoted as λ_{max} . It should be kept in mind that λ_{max} does not correspond to the maximal value of the wavelength of an isotherm taking place at temperature T of the black body. It is rather the wavelength corresponding to the peak of each isotherm. For example, for the isotherm at 5,500 K, $\lambda \approx 520 \text{ nm}$.

Black Body Isotherm Curves



Rayleigh–Jeans law

The Rayleigh–Jeans law is an approximation to the spectral radiance of electromagnetic radiation as a function of wavelength from a black body at a given temperature. It can be expressed as:

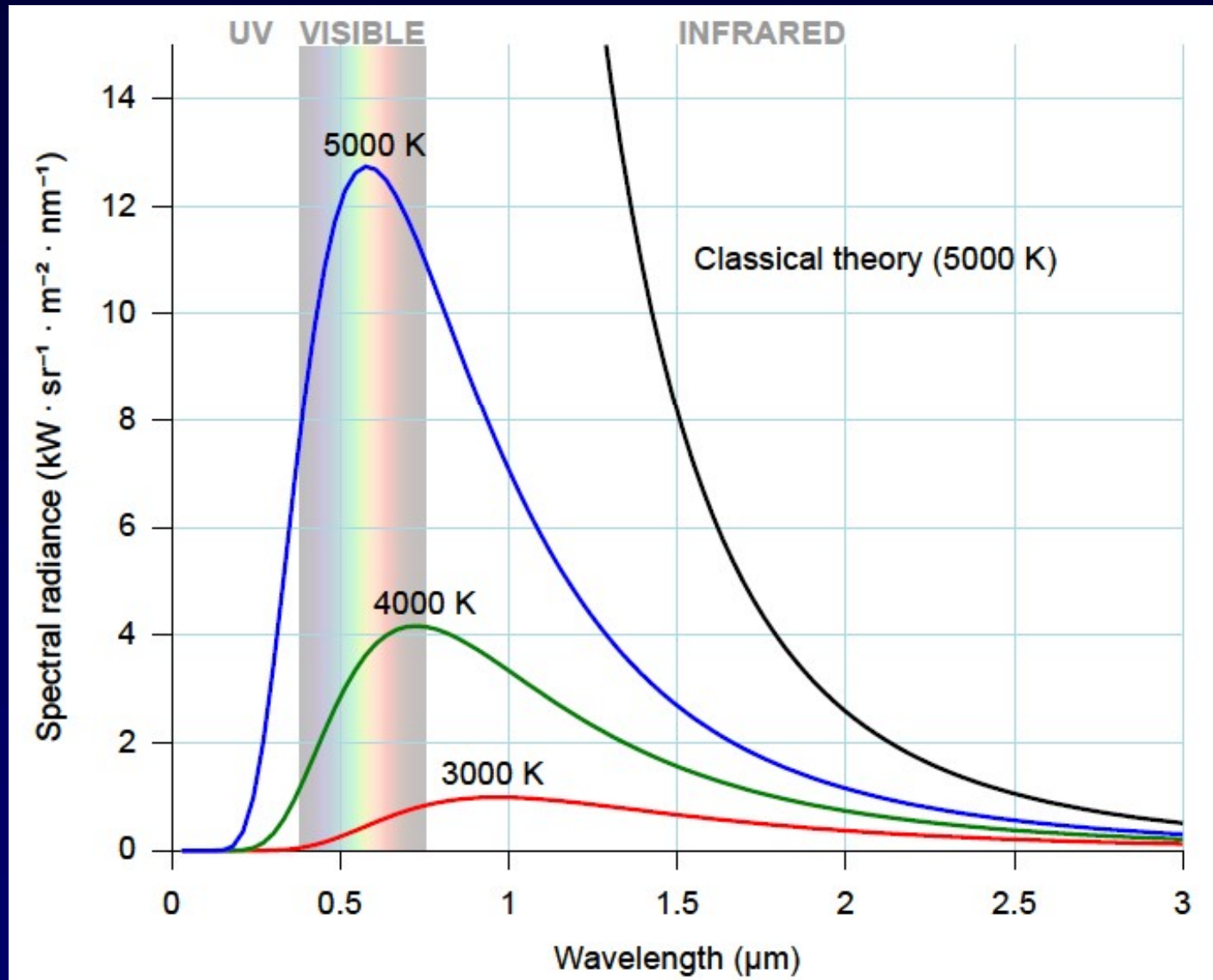
$$B_{\lambda}(T) = \frac{2ck_{\text{B}}T}{\lambda^4},$$

where B_{λ} is the spectral radiance, the power emitted per unit emitting area, per steradian, per unit wavelength; c is the speed of light; k_{B} is the Boltzmann constant; and T is the temperature in kelvins. For frequency ν , the expression is:

$$B_{\nu}(T) = \frac{2\nu^2 k_{\text{B}}T}{c^2}$$

The Rayleigh–Jeans law agrees with experimental results at large wavelengths (low frequencies) but strongly disagrees at short wavelengths (high frequencies).

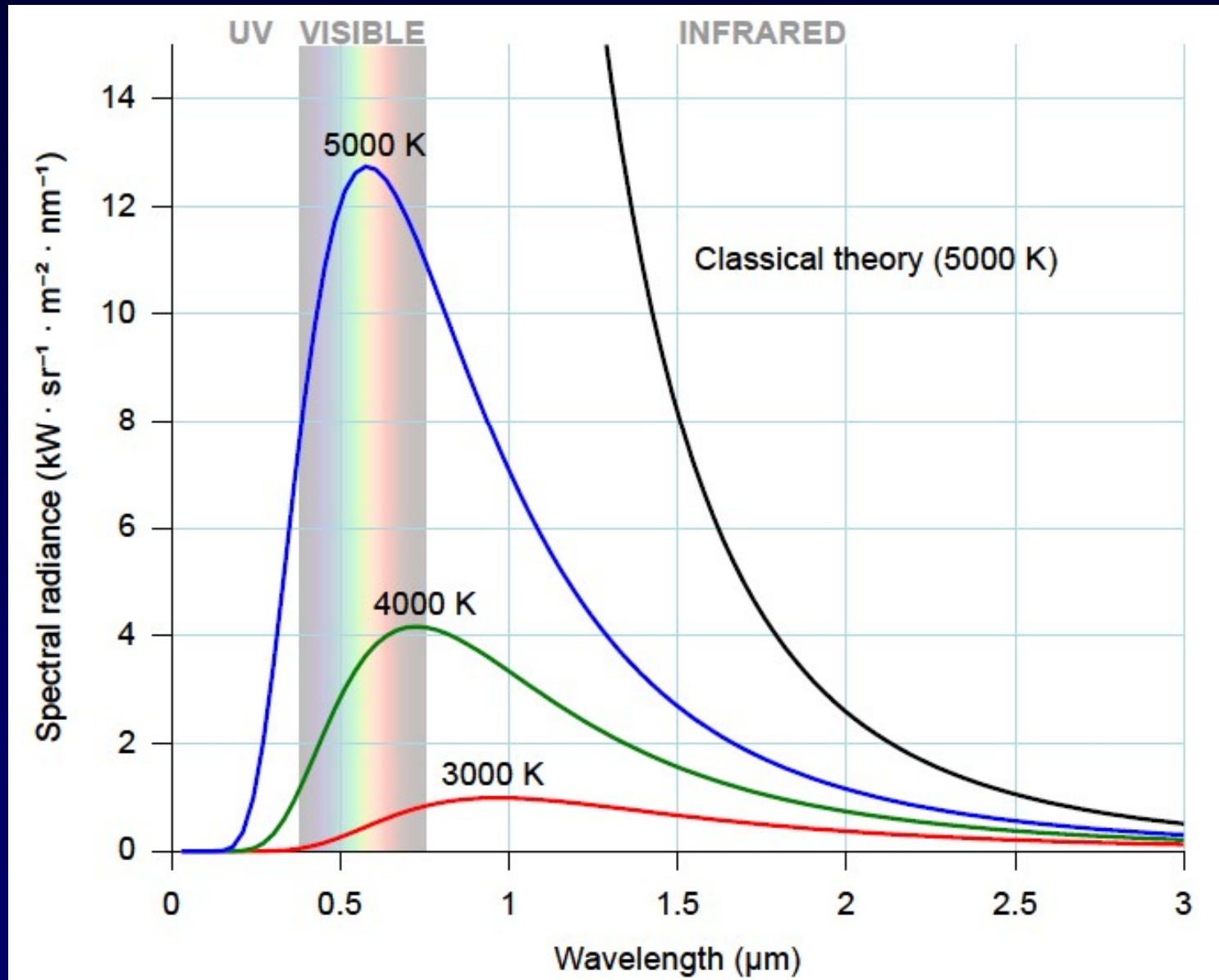
Spectral radiance of electromagnetic radiation Following Rayleigh–Jeans law and Plank's law



Ultraviolet Catastrophe / Rayleigh - Jeans catastrophe

- The ultraviolet catastrophe, also called the Rayleigh - Jeans catastrophe, was the prediction of late 19th century/early 20th century.
- An ideal black body at thermal equilibrium would emit unbound quantity of energy as wavelength decreases.
- Infact, the ultraviolet catastrophe is the error at short wavelengths in the Rayleigh - Jeans law (depicted as "classical theory" in the graph) for the energy emitted by an ideal black body. The error, much more pronounced for short wavelengths, is the difference between the black curve (as classically predicted by the Rayleigh - Jeans law) and the blue curve (the measured curve as predicted by Planck's law).

Ultraviolet Catastrophe / Rayleigh - Jeans catastrophe



Black-Body Radiation Laws

Kirchhoff's Laws of Thermal Radiation

- Gustav Robert Kirchhoff was a German physicist. He is well known especially for his laws related to the conservation of currents and charges in electrical circuits. Kirchhoff is also known for his laws related to thermal radiation, which he formulated in 1859.
- There are two *Kirchhoff's laws* on the thermal radiation.
- They explain the black body isotherms and the relation between radiance L of the gray body and radiance L_0 of the black body.
- **The First Law:** all black bodies at the same temperature have the same radiance.
- **The Second Law:** among all the objects brought to the same temperature, the black body is the most luminous.
- Considering a gray body of coefficient of absorption α , the mathematical expression of Kirchhoff's second law leads to the relation between radiance L of the gray body and radiance L_0 of the black body, which is:

$$L = \alpha \cdot L_0$$

Black-Body Radiation Laws

The Stefan-Boltzmann Law

- **Joseph Stefan** was an Austrian physicist. He is especially renowned for his work published in 1879 on the radiation of the black body in which he stated the law that bears his name.
- Based on this law, Stefan determined the Sun's surface temperature (5430°C).
- Then his student Boltzmann offered a theoretical justification for the Stefan law. This is why this law is commonly known as the Stefan - Boltzmann law.

Black-Body Radiation Laws

The Stefan-Boltzmann Law

By definition, total energy exitance M^0 is equal to the power radiated by the unit surface in all directions and can be computed using:

$$M^0 = \sigma T^4 \text{ ----- (1)}$$

Where, σ is Stefan - Boltzmann constant which can be written as $\sigma = 5.67 * 10^{-8} \text{ J s}^{-1}\text{m}^{-2}\text{K}^{-4}$, and L_0 is the radiance of the black body.

The total radiant exitance for the gray body (M) and black body (M^0) are related to radiance L (gray body) and L_0 (black body) as per following:

$$M = \pi L \quad \text{and} \quad M^0 = \pi L_0 \text{ ----- (2)}$$

Therefore the Stefan - Boltzmann law states that the total radiant exitance M^0 of the black body is proportional to the fourth power of its temperature T .

Using above two relations (1) and (2) , the radiance L_0 of the black body can be written as:

$$L_0 = \sigma T^4 / \pi \text{ ----- (3)}$$

Black-Body Radiation Laws

Wien's laws and Useful Spectrum

The two Wien's laws give the abscissa $\lambda_{\max}$ of the wavelength and the ordinate $M_{\lambda_{\max}}^0$ of the maximum *monochromatic exitance* for each temperature T of the black body.

Wien's first law (Wien's Displacement Law)

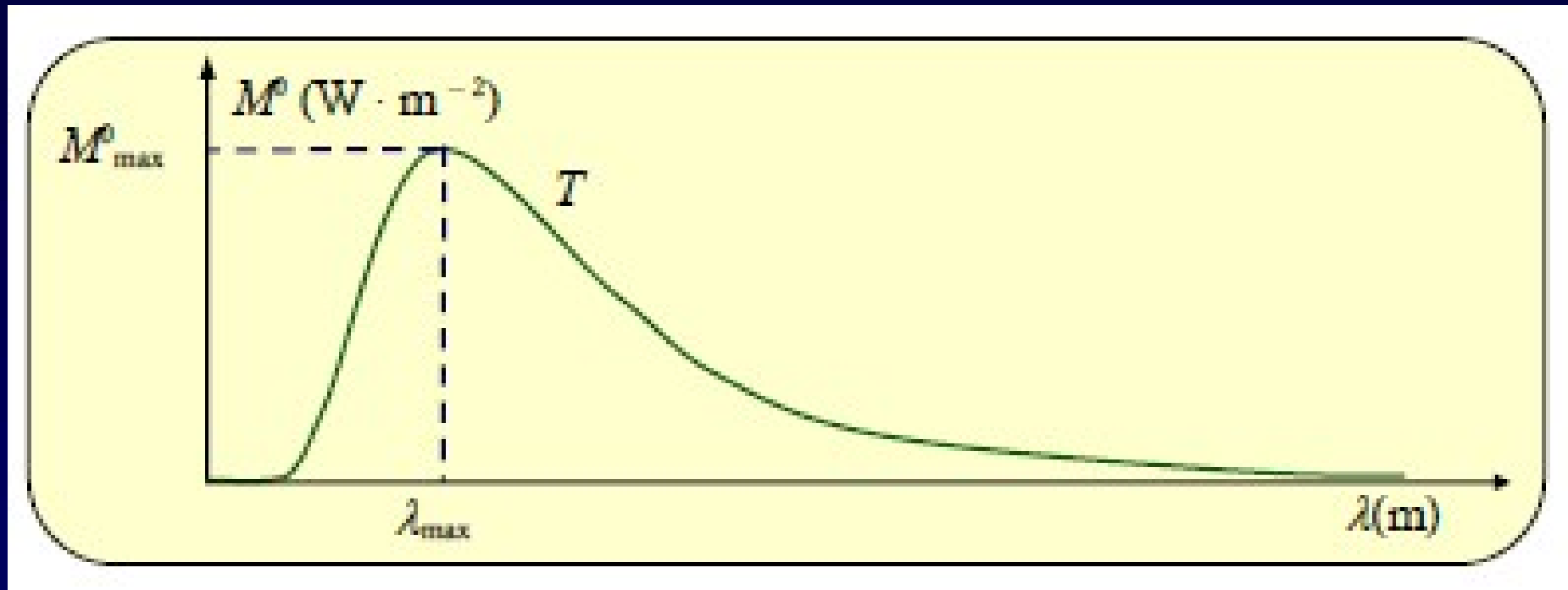
The wavelength $\lambda_{\max}$ at an isotherm peak shifts toward short wavelengths when temperature T increases. According to this law:

$$\lambda_{\max} T = \sigma_w$$

Where, σ_w is a constant, known as Wien's displacement constant, which is $\sigma_w = 2.8977685 \times 10^{-3} \text{ m} \cdot \text{K} = 2.8977685 \times 10^6 \text{ nm} \cdot \text{K}$.

Wien's law shows that the value of $\lambda_{\max}$ shifts toward short wavelengths when the temperature T increases.

Isotherm curve of the black body at temperature T



$$\lambda_{\text{max}} T = \sigma_w$$

Wien's second law

The ordinate $M_{\lambda_{\max}}^0$ of the maximum monochromatic exitance is proportional to the fifth power of temperature, which is:

$$M_{\lambda_{\max}}^0 = BT^5$$

In the above relation, $M_{\lambda_{\max}}^0$ is expressed in $W \cdot m^{-3}$ (if the wavelength $\lambda_{\max}$ is expressed in m) or in $W \cdot m^{-2} \cdot \mu m^{-1}$ (if $\lambda_{\max}$ is expressed in micrometers); the units of constant B depend on the unit of wavelength $\lambda_{\max}$.

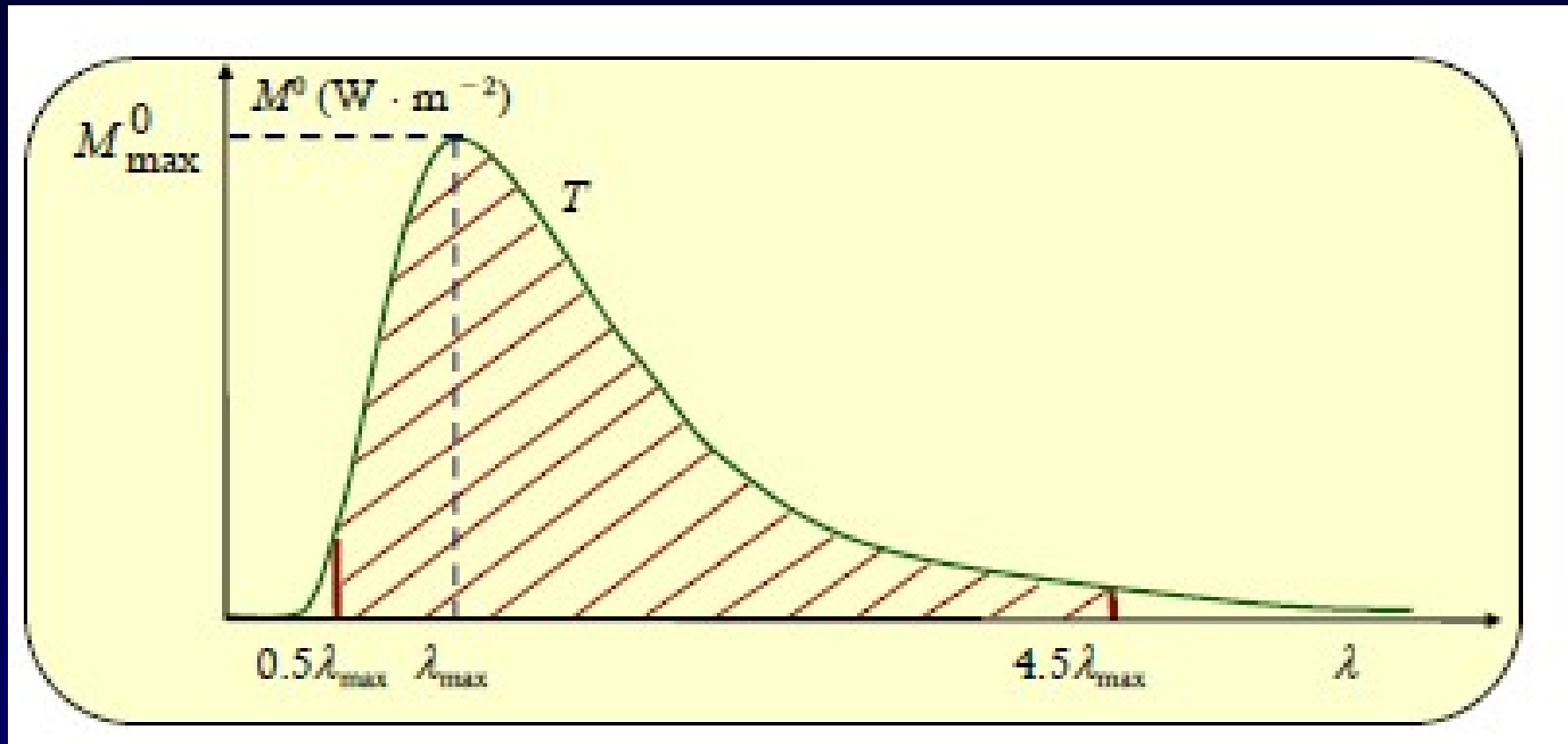
$B = 1.28 \cdot 10^{-5} [W \cdot m^{-3} \cdot K^{-5}]$ if $\lambda_{\max}$ is expressed in m.

$B = 1.28 \cdot 10^{-11} [W \cdot m^{-2} \cdot \mu m^{-1} \cdot K^{-5}]$ if $\lambda_{\max}$ is expressed in μm .

It is noticed that when the wavelength of radiation is such that $\lambda < 0.5\lambda_{\max}$, there is practically no more radiated energy (approximately 1%).

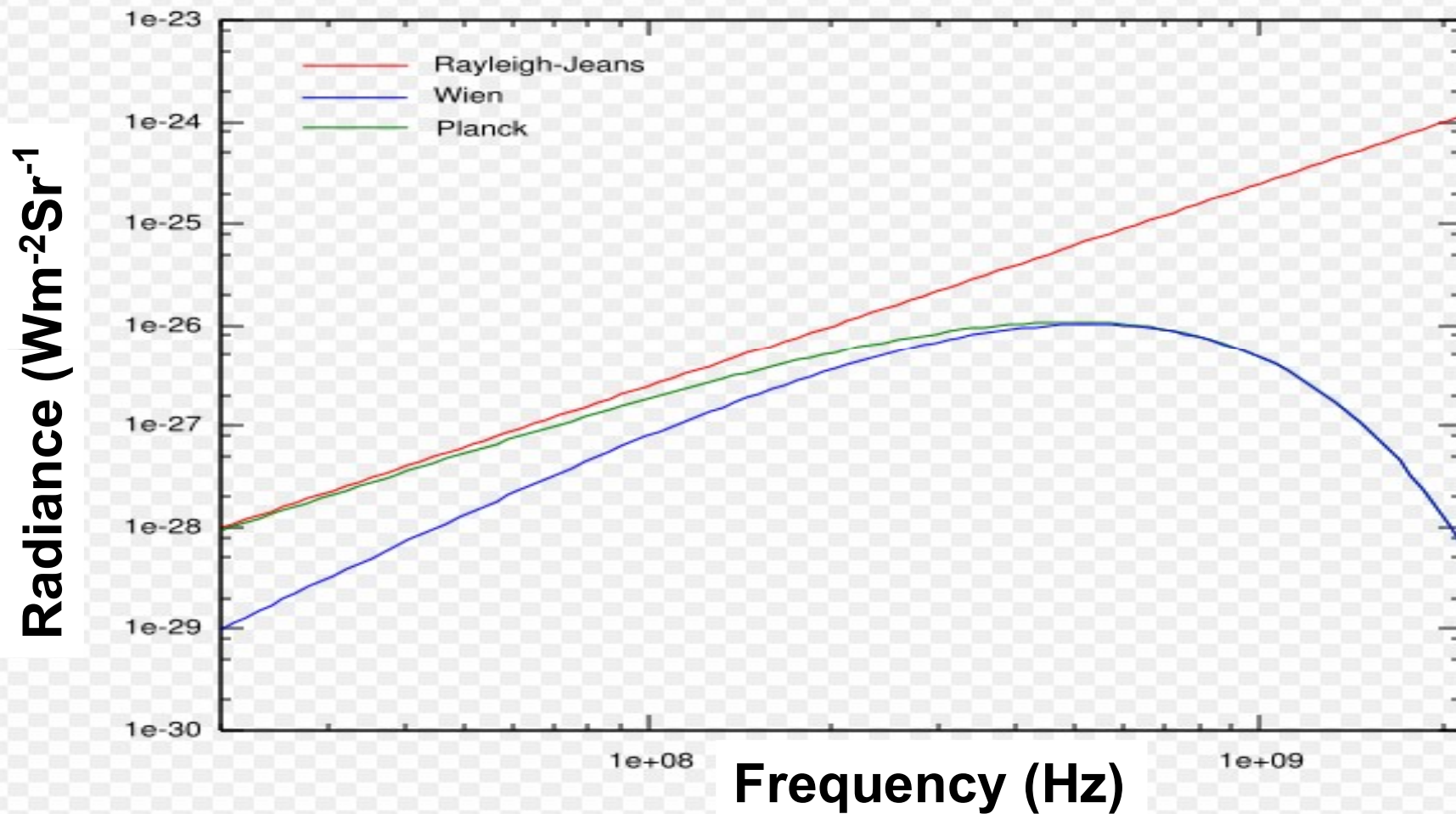
Furthermore, there is practically no more radiated energy when $\lambda > 4.5\lambda_{\max}$. By definition, the wavelength range $0.5\lambda_{\max} < \lambda < 4.5\lambda_{\max}$ is called the **useful spectrum** of the considered isotherm (hatched part in Figure).

Useful Spectrum (Hatched Part of the Isotherm Curve)



The wavelength range $0.5\lambda_{\text{max}} < \lambda < 4.5\lambda_{\text{max}}$ is called the useful spectrum of the considered isotherm (hatched portion in the figure).

Comparison of Rayleigh-Jeans, Wien's and Planck's laws for a Body of 8 mK Temperature



Summary of Black-Body Properties

It has been shown that the blackbody possesses certain fundamental properties that make it a useful standard for quantifying the properties of real radiating bodies. These properties are the following:

1. The blackbody is the maximum absorber and emitter of radiant energy at all wavelengths and in all directions. Therefore it reflects no radiation and appears perfectly black.
2. Roughly we can say that the stars radiate like blackbody radiators. This is important because it means that we can use the theory for blackbody radiators to infer important properties about stars.
3. At a particular temperature the black body would emit the maximum amount of energy possible for that temperature.
4. Blackbody radiation does not depend on the type of object emitting it. Entire spectrum of blackbody radiation depends on only one parameter, the temperature, T .

Summary of Black-Body Properties

5. The total (including all wavelengths) radiant exitance (total emissive power) of a blackbody into a medium is *given by the Stefan–Boltzmann law*:

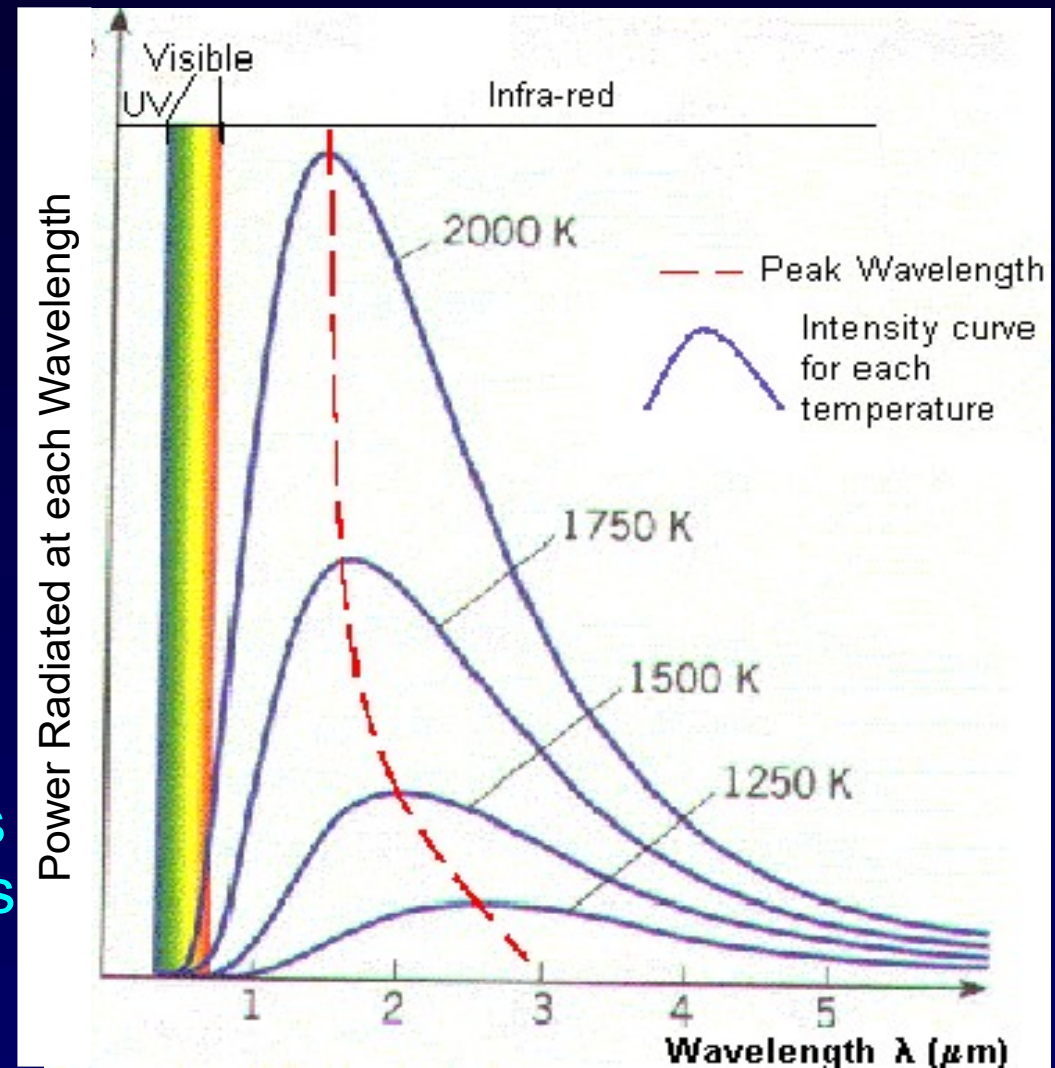
$$M^0 = \sigma T^4$$

6. The blackbody spectral and total intensities are independent of the direction so that emission of energy into a direction θ away from the surface normal direction is proportional to the projected area of the emitting element, $dA \cos\theta$. This is known as Lambert's cosine law.
7. Planck's law describes the electromagnetic radiation emitted by a black body in thermal equilibrium at a definite temperature.
8. The wavelength at which the maximum spectral intensity of radiation into a medium for a blackbody is given by Wien's displacement law.

$$\lambda_{\max} T = \sigma_w$$

9.

- As the temperature increases, the peak wavelength emitted by the black body decreases.
- As temperature increases, the total energy emitted increases, because the total area under the curve increases.
- As the temperature decreases, the curve gets infinitely close to the x-axis but never touches it.



Numerical - 1

Consider an isotherm at the surface temperature of the Sun, which is assimilated to a black body. Find the useful spectrum corresponding to $T = 6,000$ K, if $\sigma_W = 2.898 \cdot 10^{-3} \text{ m} \cdot \text{K}$.

Solution:

$$\lambda_{\max} \cdot T = \sigma_W$$

$$\begin{aligned}\lambda_{\max} &= (2.898 \times 10^{-3} \text{ m} \cdot \text{K}) / 6000 \text{ K} \\ &= 0.483 \times 10^{-6} \text{ m} \\ &= 483 \times 10^{-9} \text{ m} \\ &= 483 \text{ nm}\end{aligned}$$

Since the wavelength range $0.5\lambda_{\max} < \lambda < 4.5\lambda_{\max}$ is called the useful spectrum of the considered isotherm.

Therefore useful spectrum corresponding to 6000 K will be:

$$\begin{aligned}0.5 \times 483 < \lambda < 4.5 \times 483 \text{ nm} \\ \text{or, } 241.5 < \lambda < 2173.5 \text{ nm}\end{aligned}$$

Numerical - 2

What is the maximal monochromatic exitance of the isotherm of the black body at wavelength $\lambda_{\max} = 1.184 \text{ } \mu\text{m}$ at $T = 2,500 \text{ K}$?
 $B = 1.28 \cdot 10^{-11} \text{ W} \cdot \text{m}^{-2} \cdot \mu\text{m}^{-1} \cdot \text{K}^{-5}$.

Solution:

The *maximum monochromatic exitance*, $M_{\lambda_{\max}}^0$, is proportional to the fifth power of temperature, which is:

$$\begin{aligned} M_{\lambda_{\max}}^0 &= BT^5 \\ &= 1.28 \cdot 10^{-11} \cdot (2500)^5 \\ &= 1.25 \cdot 10^6 \text{ Wm}^{-2}\mu\text{m}^{-1} \end{aligned}$$

Climate Forcing

Weather and Climate

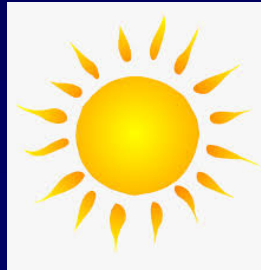
- **Weather** - refers to the state of the atmosphere at a given place and time.
- **Climate** - Climate is often summarized by average or mean values of weather elements, but it is necessary to include the extremes and variations. Climate represents for a larger area.

Weather Vs Climate

Weather



Can change within a few minutes or hours



Climate



Takes very long time to change!

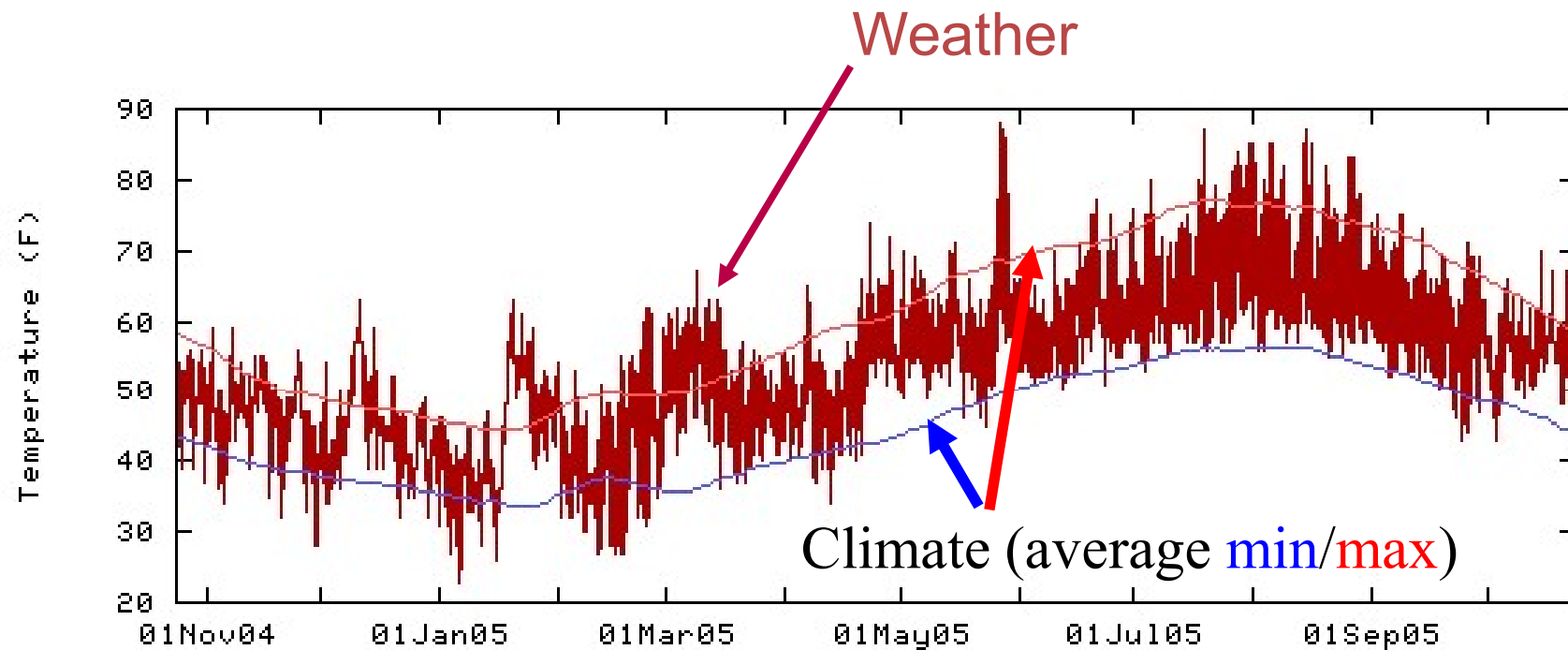


Climate Vs Weather

- **Determined: Over time**
- **Looked at by the years**
- **Reported as an average**
- **Depends on the location on Earth**
- **Weather makes up climate**

- **Determined: daily**
- **Looked at by the minutes, hour, day, week**
- **Reported as a forecast**
- **Depends on the weather occurring mainly to the west**
- **Climate helps you determine the weather**

Climate is an average of weather (Temperature, Rainfall...) over a “long” time.

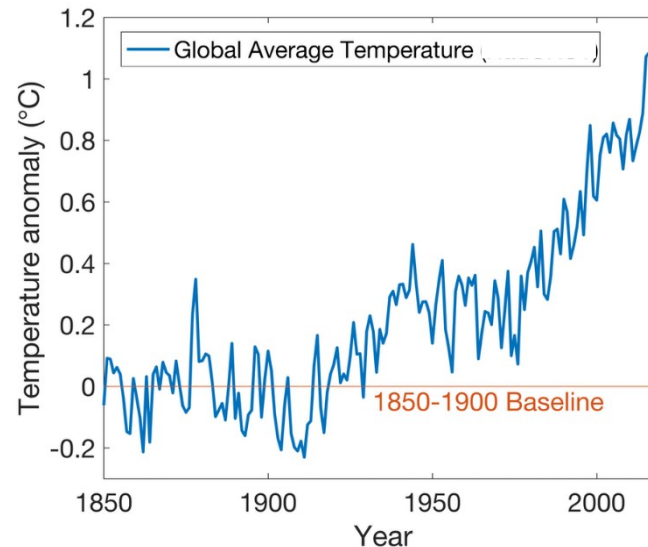
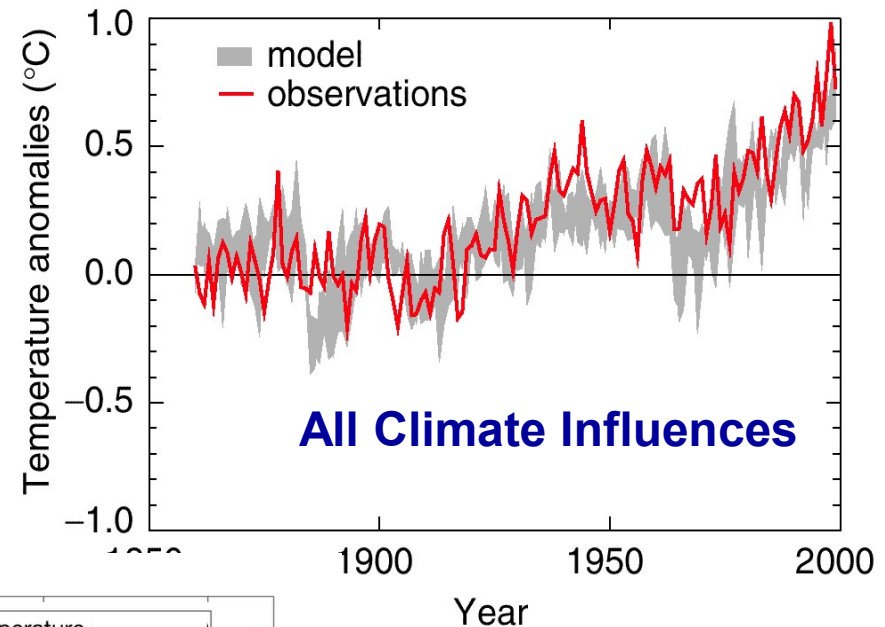
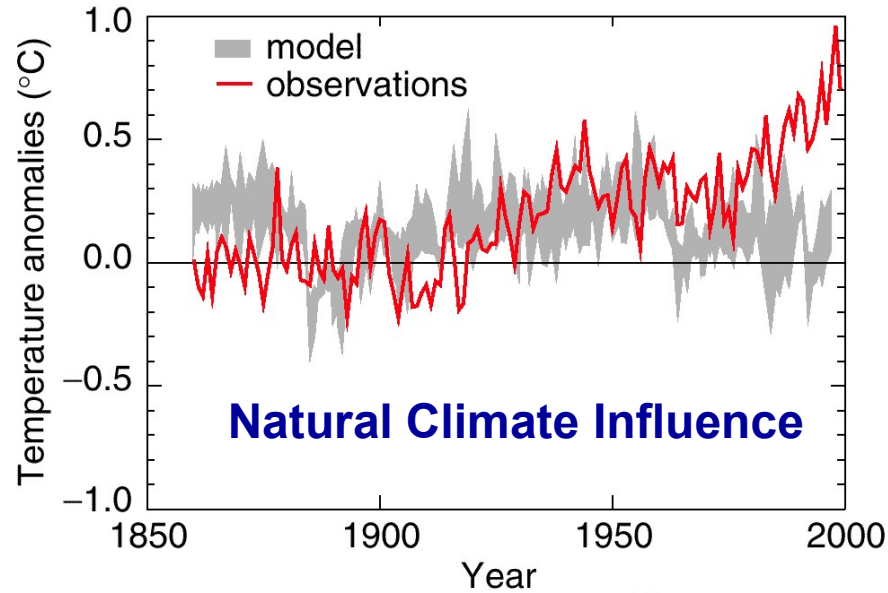


Climate Forcing

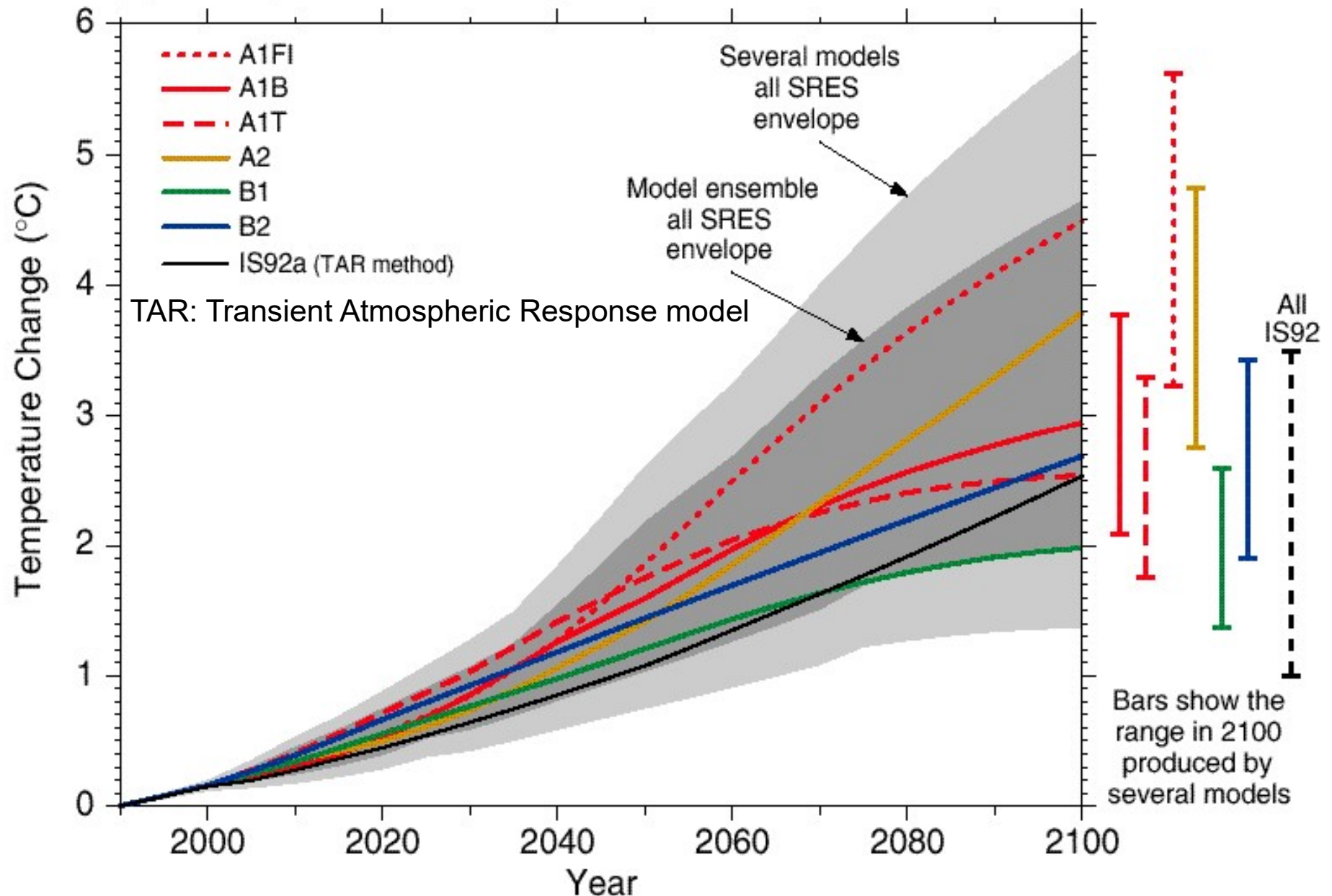
- Climate forcing is the physical process of affecting the climate on the Earth through a number of forcing factors.
- These factors are specifically known as forcings because they drive the climate to change, and it is important to note that these forcings exist outside the existing climate system.
- The climate system includes the hydrosphere, land surface, the cryosphere, the biosphere, and atmosphere.
- Examples of some of the most important types of forcings include: variations in solar radiation, volcanic eruptions, changing albedo, and changing levels of greenhouse gases in the atmosphere.
- Each of these are considered external forcings because these events change independently of the climate, perhaps as a result of changes in solar activity or human-caused fossil fuel combustion.

Have we changed the climate?

Climate models capture the last 100 year's warming only if we include the human effects.



What will the future climate look like?



Intergovernmental Panel on Climate Change (IPCC) (www.ipcc.ch)

What makes climate to change?

Natural processes

- Changes in the Insolation
- Changes in the earth's orbit
- Changes in the atmosphere
- Changes in ice sheets
- Volcanic eruptions
- Tectonic Plate Movement



Human Activities

- Any activity that releases “greenhouse gases” into the atmosphere

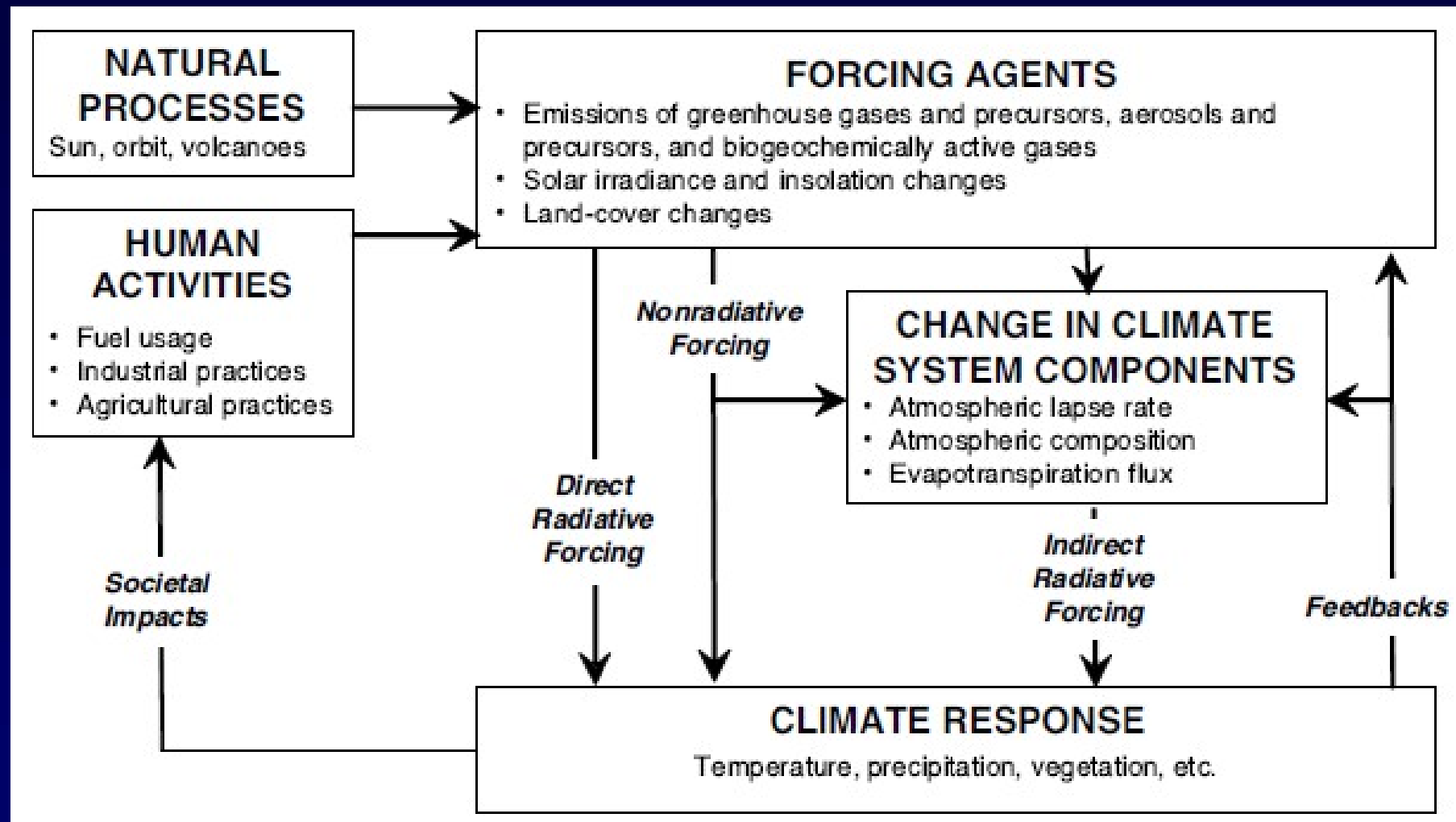
Internal Wiggles (for example El Nino)

Some cause *bigger* changes, some cause *small* changes

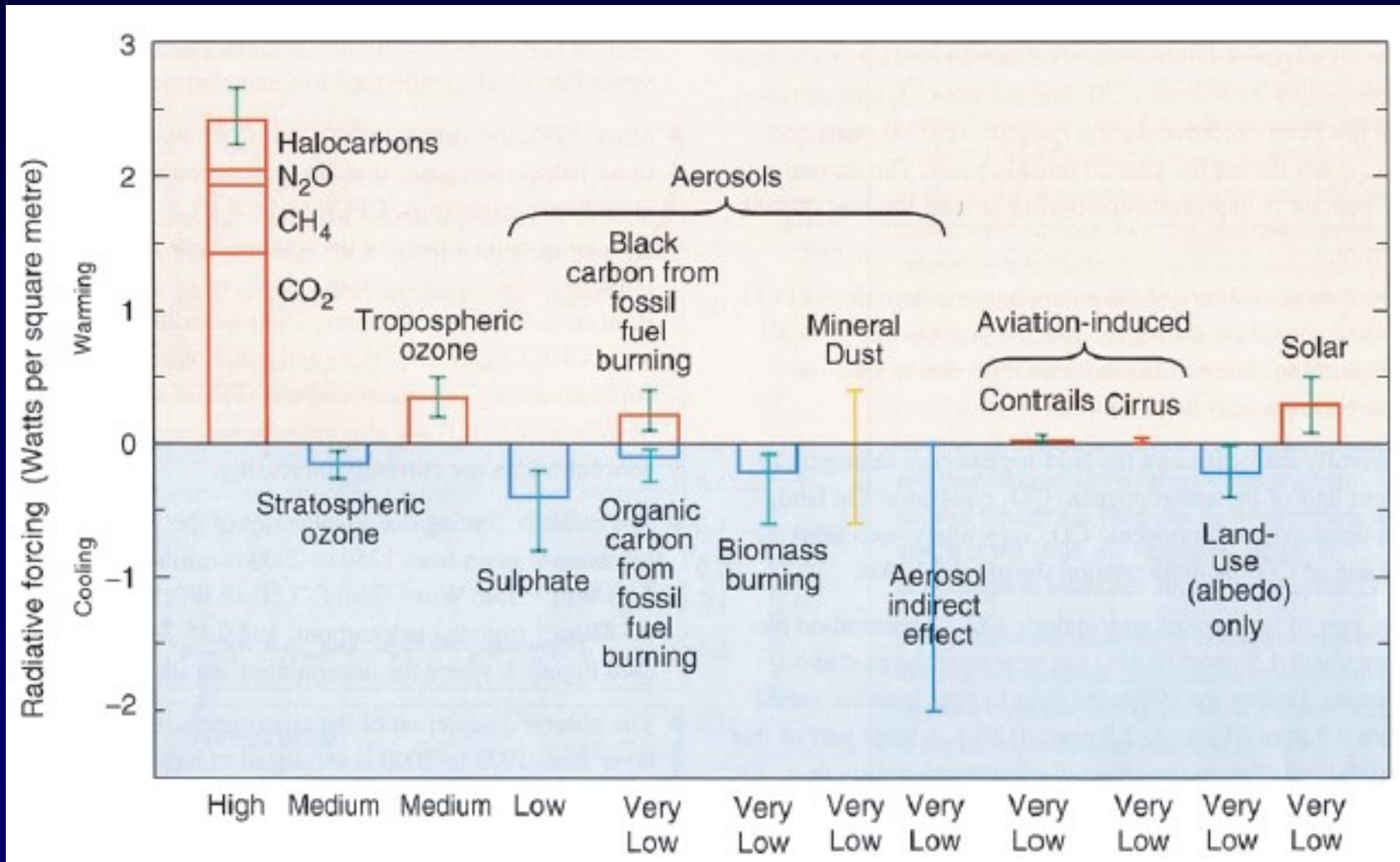
Some cause *slower* changes, some cause *fast* changes

Some cause changes that *last*, some cause changes that *go away fast*

Conceptual framework of climate forcing, response, and feedbacks



Estimated radiative forcings since pre-industrial times for the Earth and troposphere system

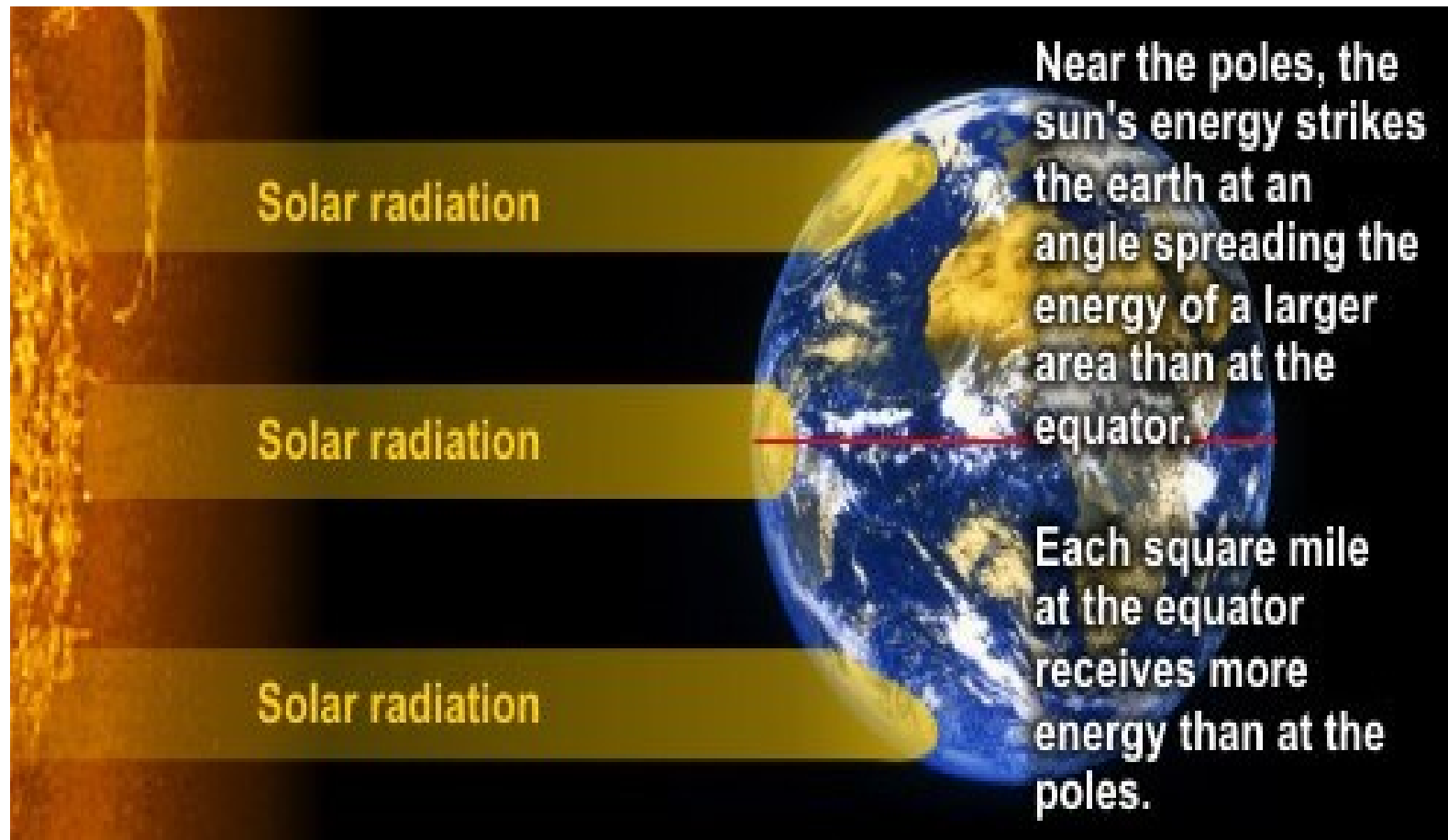


THE CLIMATE OF THE EARTH IS CONTROLLED BY THE EARTH'S ENERGY BALANCE

- Energy flows into the Earth from the Sun and flows out when it is radiated into space.
- The Earth's energy balance is determined by the amount of sun's radiation that falls on the Earth (insolation) and the characteristics of the Earth's surface and atmosphere that act to reflect, circulate and re-radiate this energy.

Insolation (Incoming **Solar radiation**): The measure of the amount of solar radiation (energy) falling on a surface. It is a very important factor in determining the climate of the Earth.

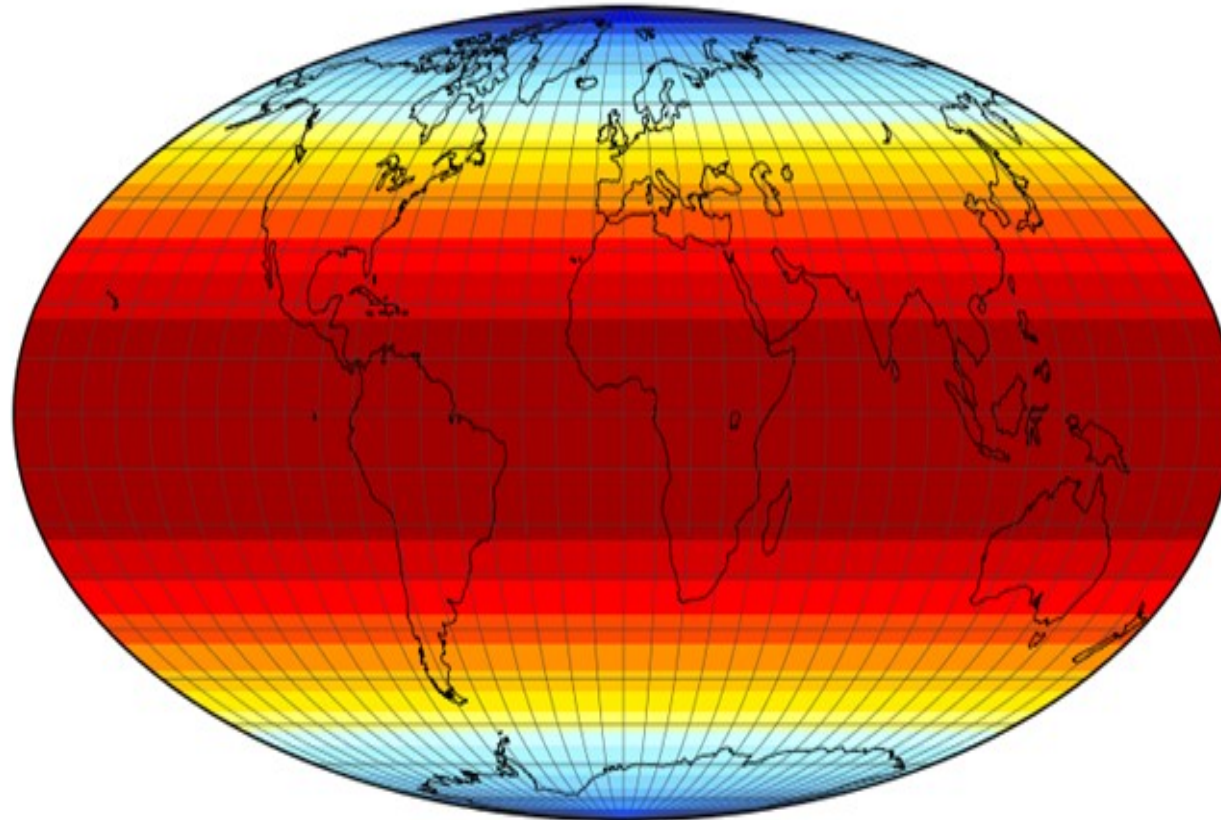
THE EFFECTS OF VARYING INSOLATION ON OUR CLIMATE



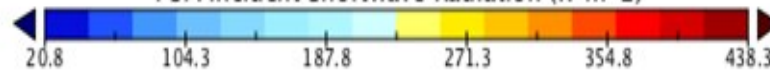
APPROXIMATELY 49% OF THE INSOLATION IS ABSORBED BY THE EARTH'S SURFACE

Top of Atmosphere Insolation

March 2003



TOA Incident Shortwave Radiation (W m^{-2})



Data Min = 20.8, Max = 438.3

DIFFERENT ORBITAL VARIATIONS AND HOW THEY EFFECT CONDITIONS ON EARTH

- The Earth's orbit around the sun is not perfectly circular
- The closer the Earth is to the Sun the more concentrated the solar radiation (insolation)

- Earth's orbit is also not fixed
- Called Milankovitch Cycles
- Changes the total insolation and the timing of the insolation

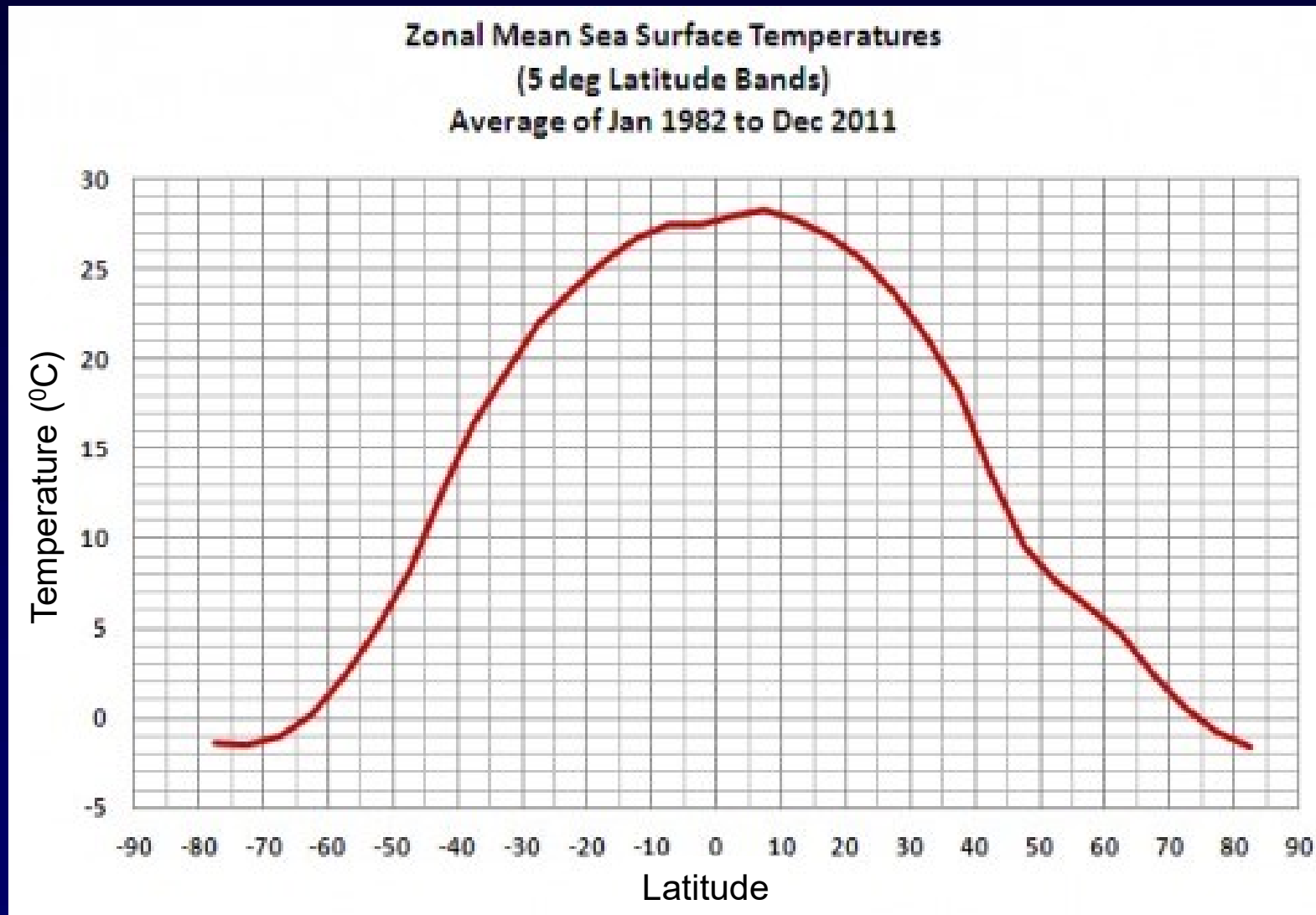
Ocean Water Warming

Factors affecting Distribution of Temperature in Ocean Water

1. Latitude
2. Inequality in the distribution of water and land
3. Prevailing wind
4. Oceanic currents
5. Distance of the earth from the sun
6. Duration of day
7. Number of sunspots activities
8. Submarine ridges
9. Local weather conditions
10. Location and shape of sea

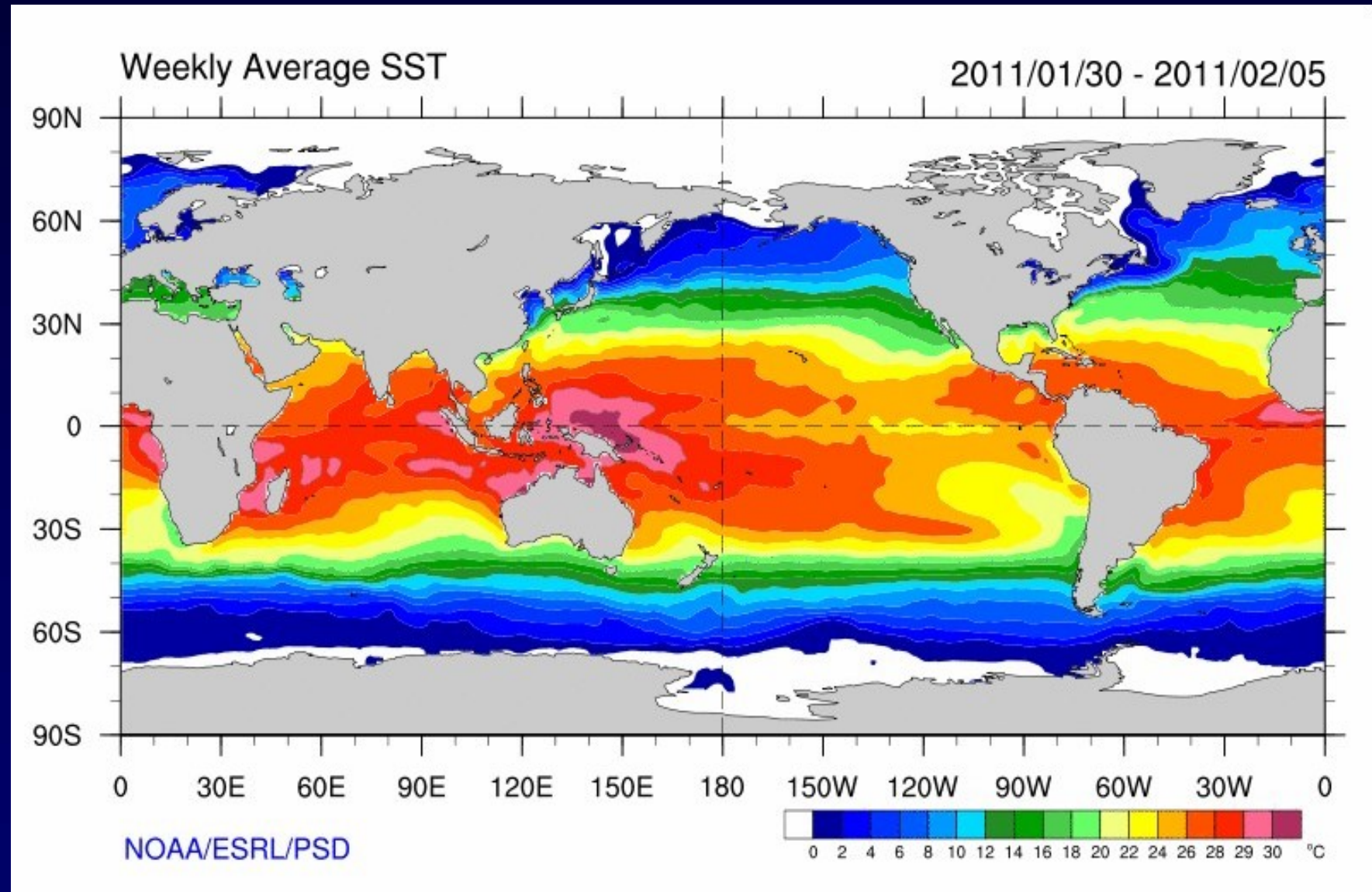
➤ Incoming Solar Radiation: Insolation

Sea Surface Temperature from Equator to Poles



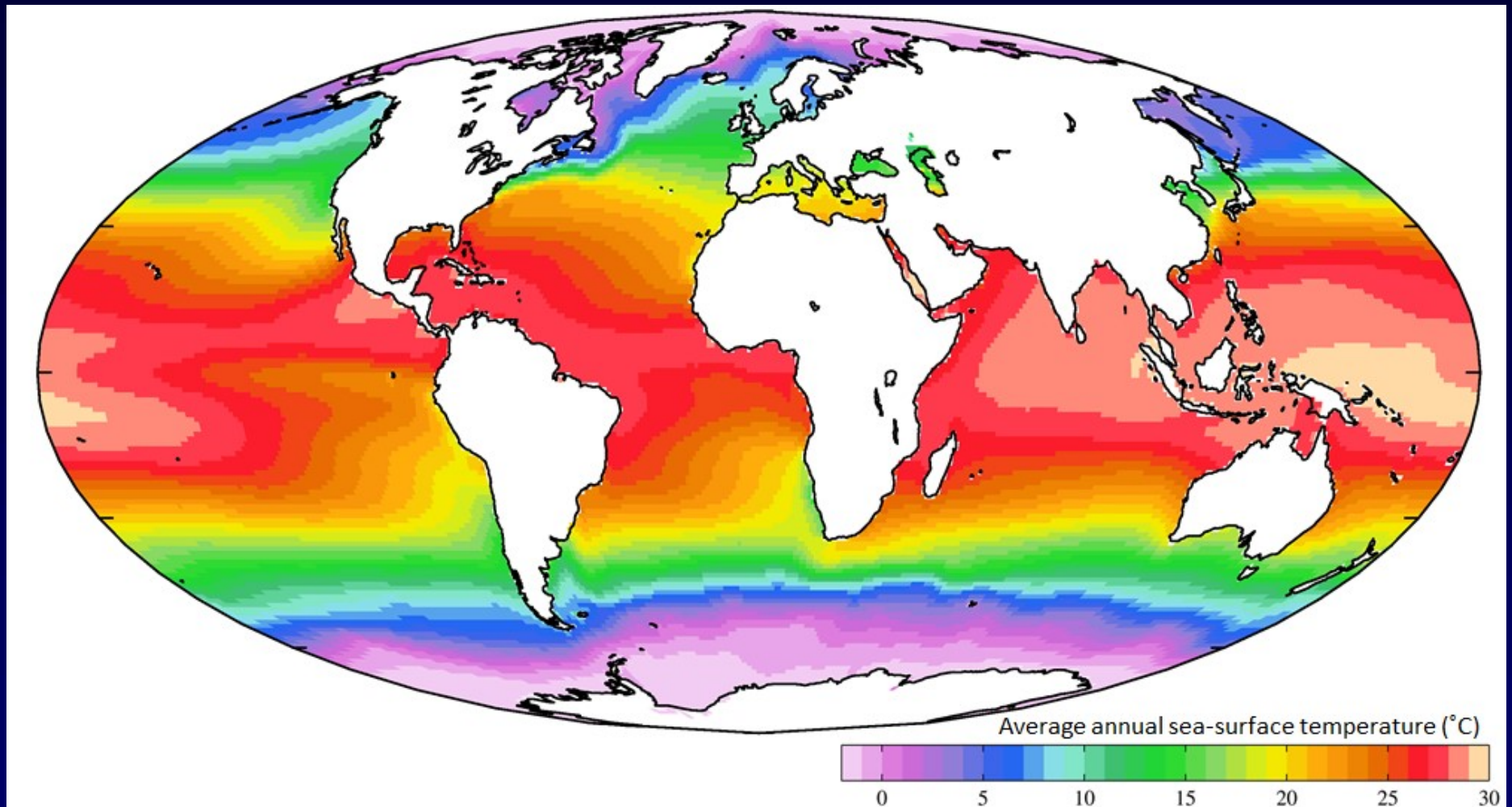
Variation with latitude of surface temperature - average of all oceans.

Global Distribution of Sea Surface Temperature



The surface temperature of seawater primarily depends on the latitude and season.

Global Average Annual Sea Surface Temperature



Vertical Profile of Ocean Temperature

➤ If thermal energy from solar radiation is largely absorbed by the surface layers, how can it be carried deeper?

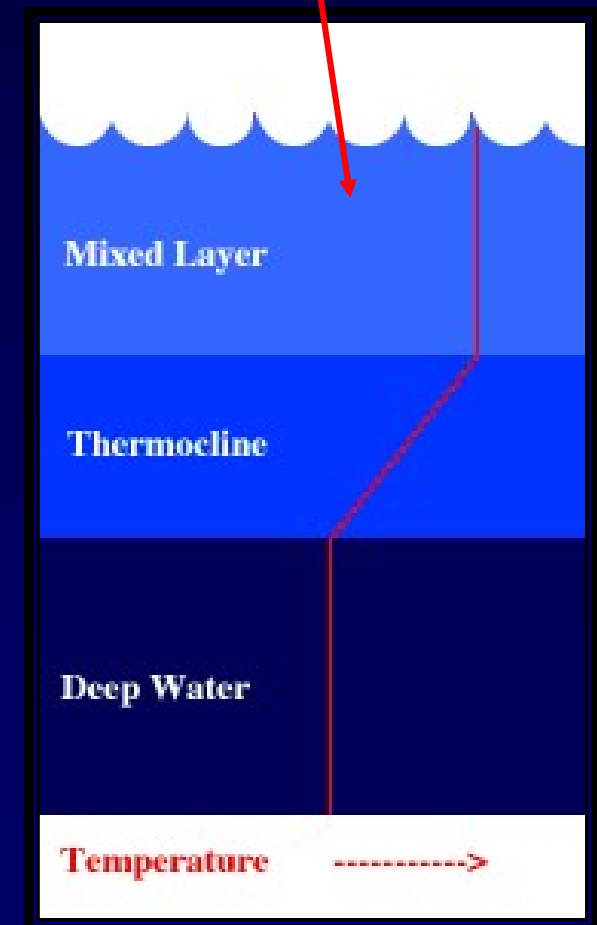
1. Convection – a slow process
2. Turbulent mixing by winds and waves
3. Horizontal and vertical advection (transport of water in bulk motion along with all its properties)



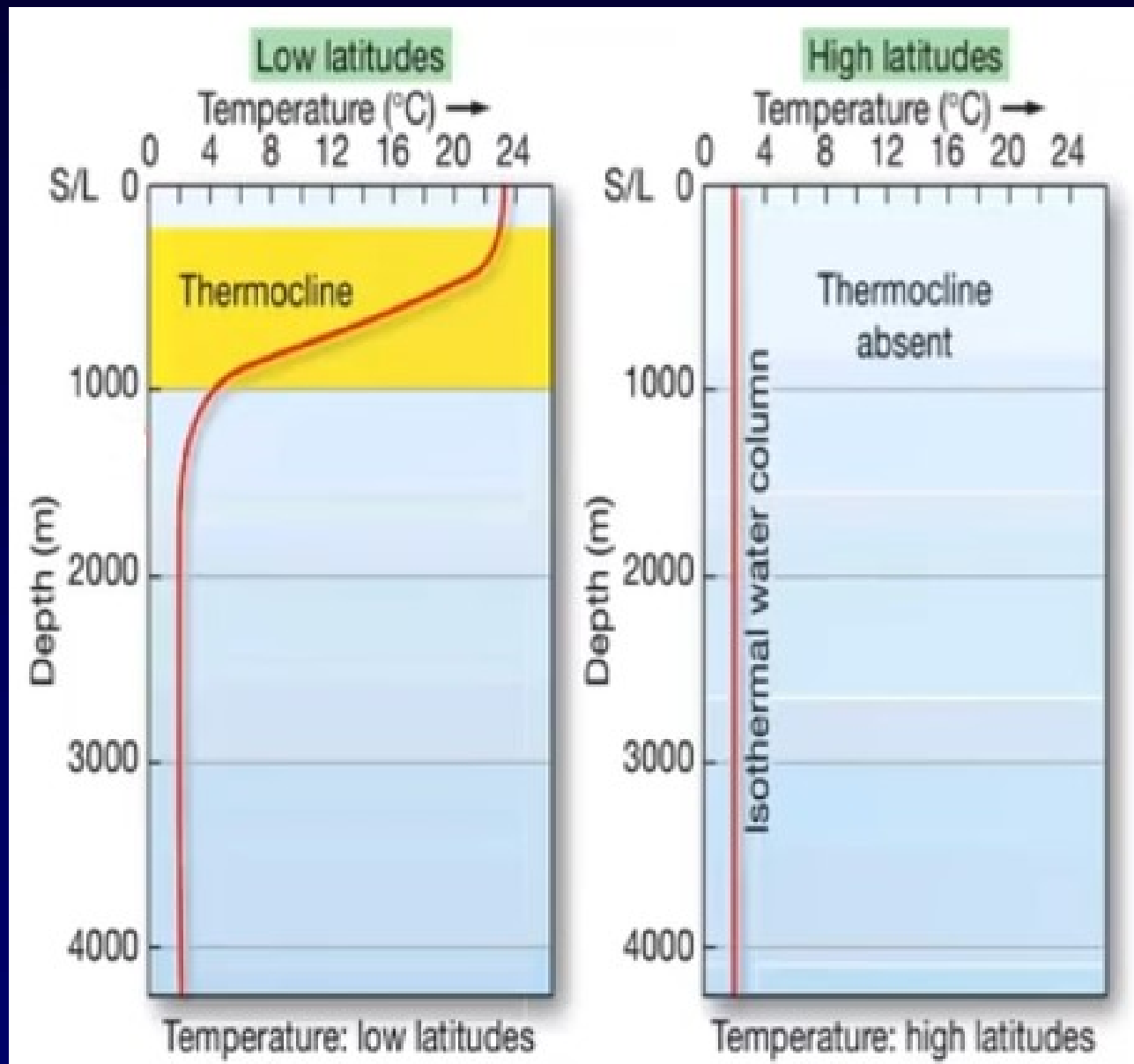
Thermal Layers of the Ocean Water

- The turbulent mixing establishes a **isothermal surface-layer** that can vary from 15 to 200 m. Depth at which Mixed layer end is called MLD (**Mixed Layer Depth**)
- Between about 200 m and 1000 m depth temperature declines rapidly. This region of steep gradient is called **Thermocline**. The thermocline is the transition layer between the mixed layer and the deep water layer. Temperature decreases sharply in this layer
- The isothermal layer and the deep water layer are relatively uniform in temperature, while the thermocline represents the transition zone between the two.

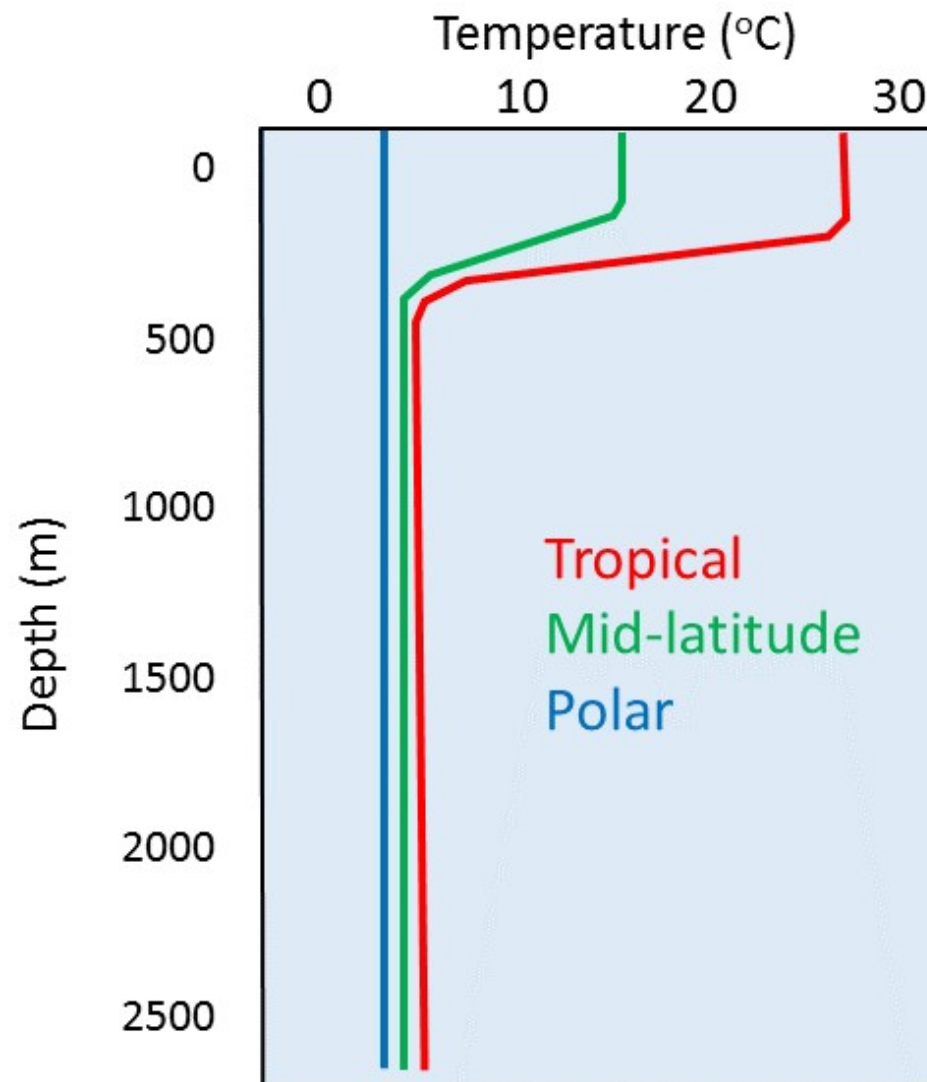
Isothermal layer



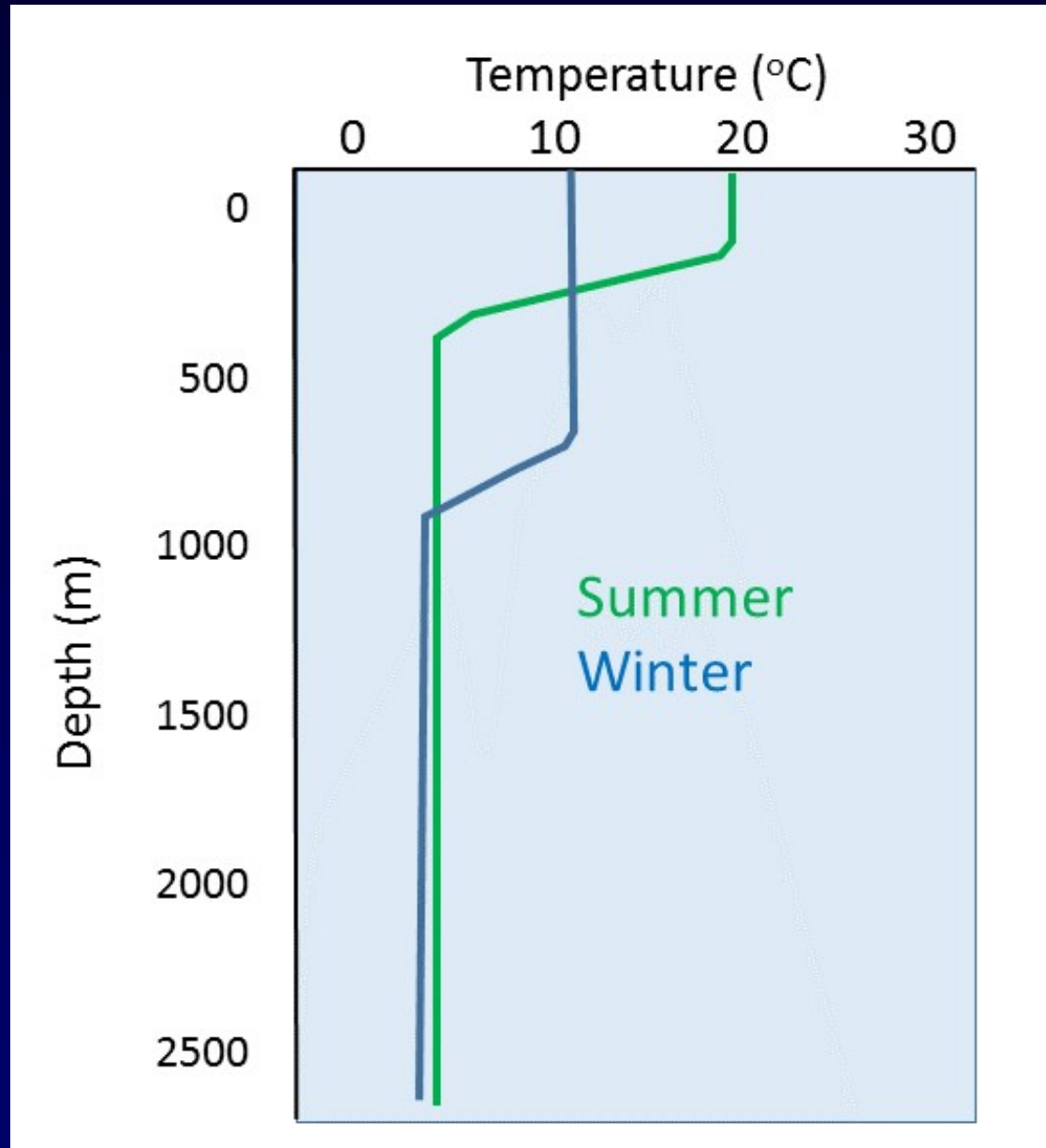
Thermocline in Low and High Latitudes



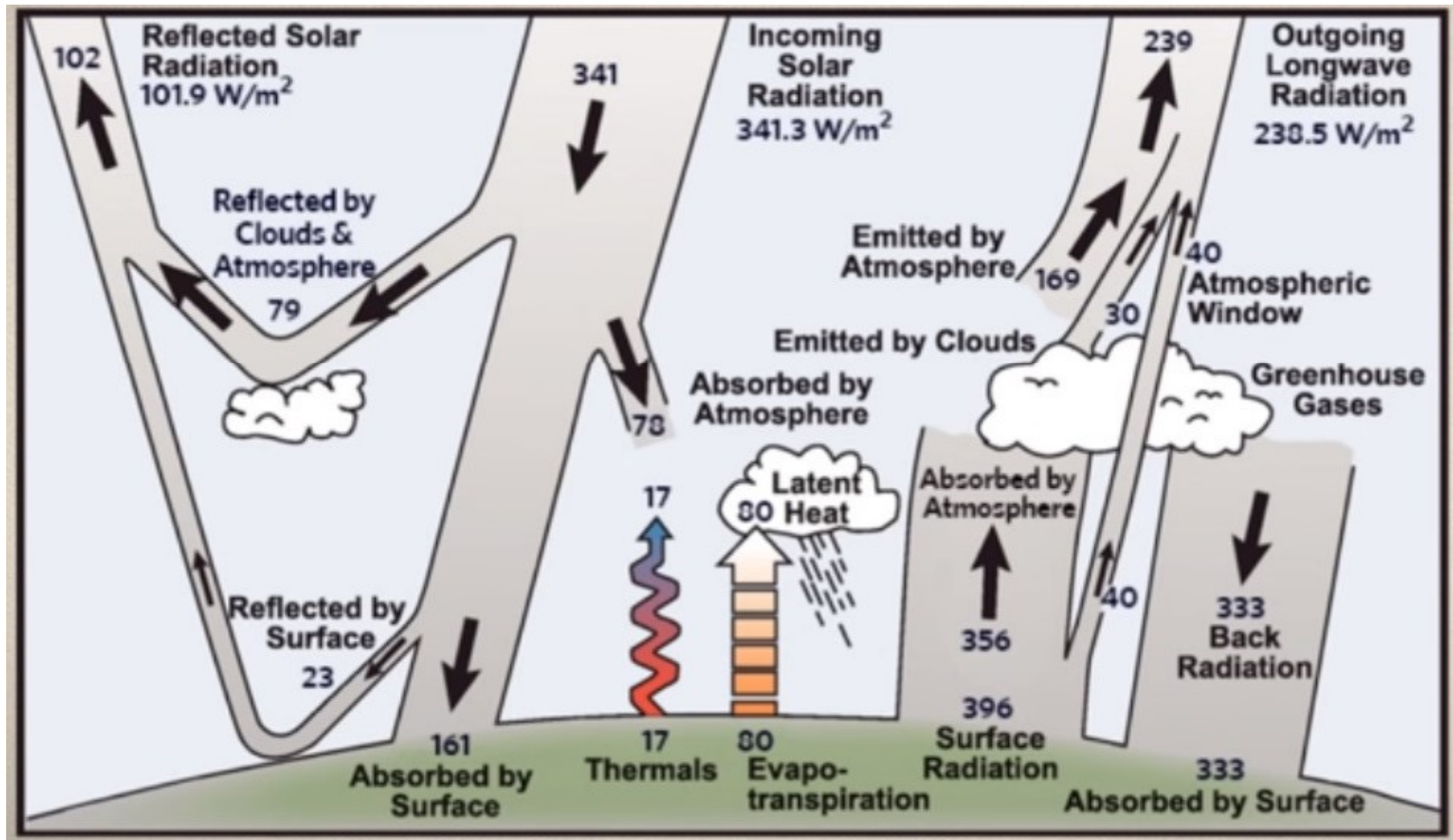
Representative Temperature Profiles for Tropical, Mid-Latitude, and Polar regions



Temperature in Summer and Winter Seasons



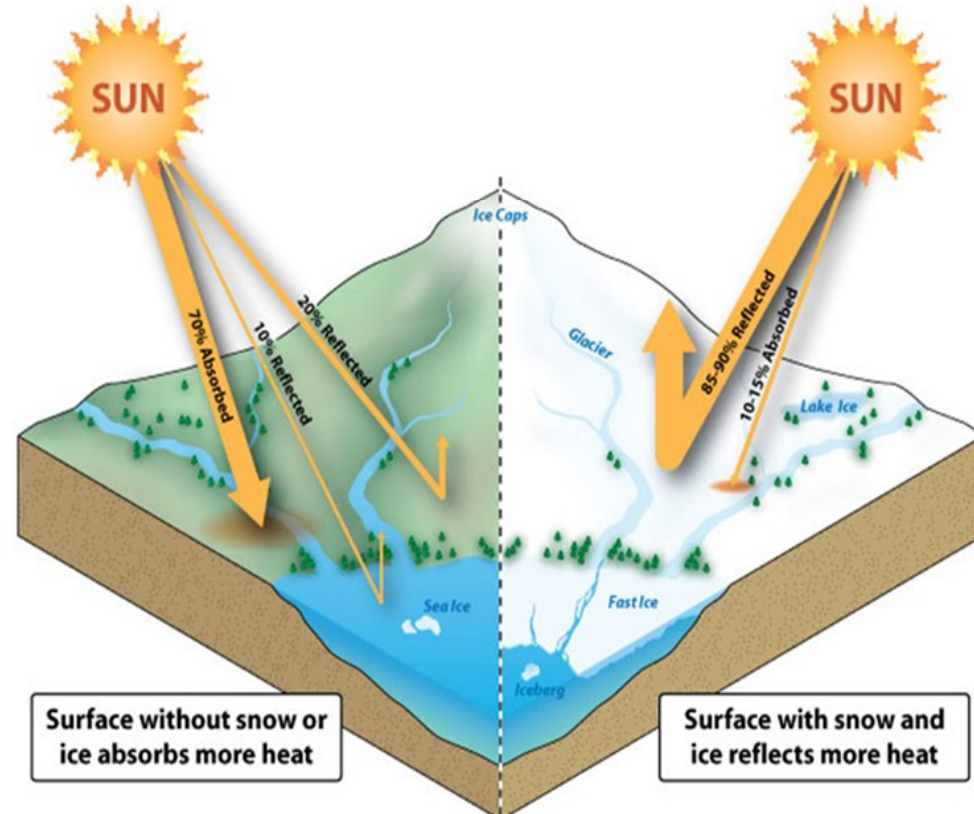
Earth's Annual Energy Budget



ALBEDO

Albedo is the fraction of light that a surface reflects. If it is all reflected, the albedo is equal to 1. If 30% is reflected, the albedo is 0.3. The albedo of Earth's surface (atmosphere, ocean, land surfaces) determines how much incoming solar energy, or light, is immediately reflected back to space. This can have an impact on climate.

The ratio of radiosity to irradiance (flux each unit of area) obtained by a surface is known as surface albedo.

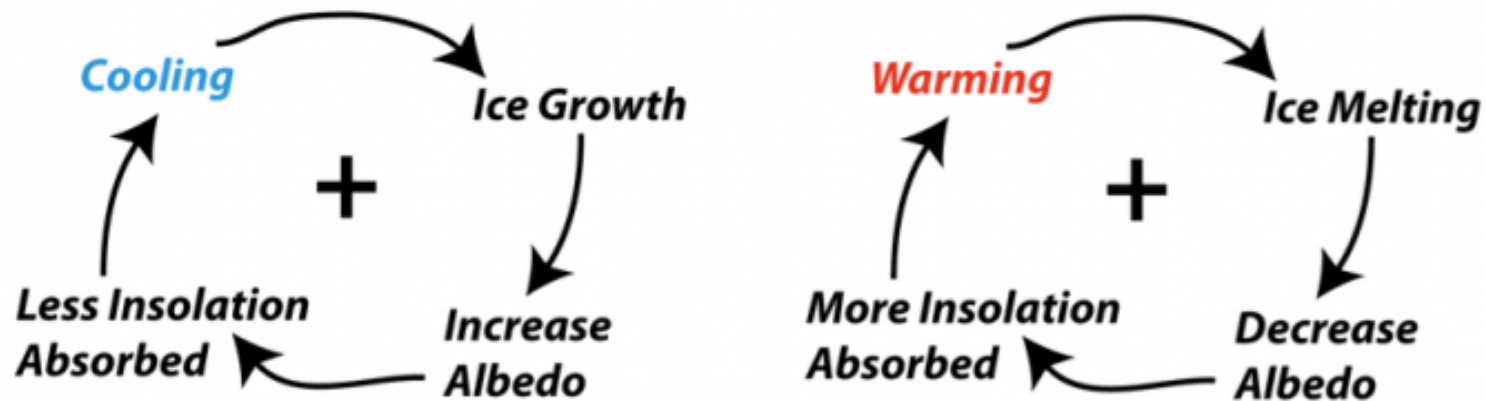


ENERGY ABSORBED VERSUS REFLECTED

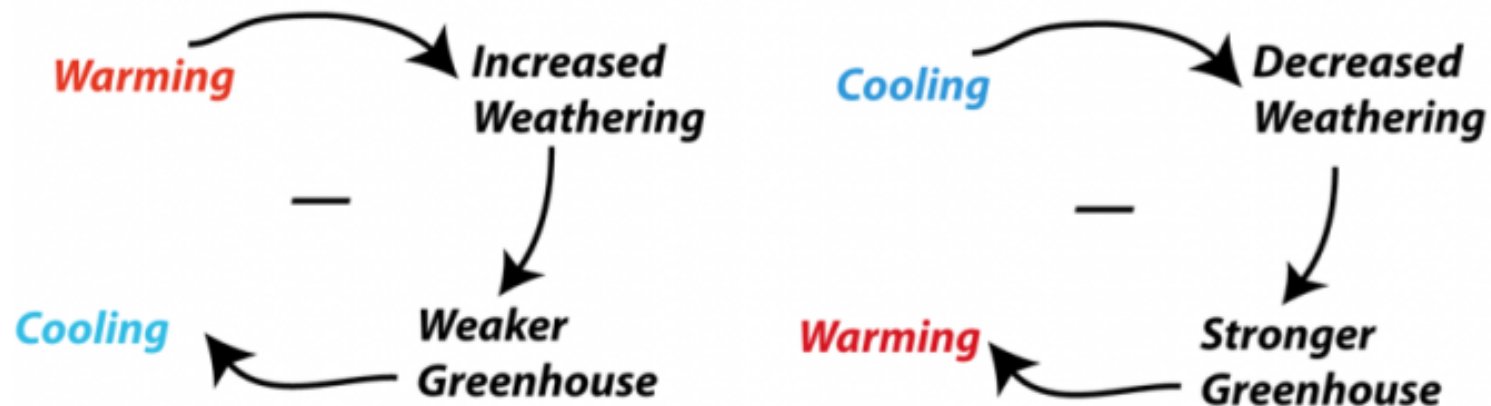
Climate change feedbacks

- **Climate change feedbacks are important in the understanding global warming.**
- **The feedback processes amplify or diminish effect of each climate forcing, and so play an important part in determining the climate sensitivity and future climate state.**
- **There are a number of feedback processes important to Earth's climate system. These are:**
 - (i) Water Vapour feedback**
 - (ii) Cloud feedback**
 - (iii) Permafrost feedback**
 - (iv) Ice-Albedo feedback**

Positive Feedback Mechanism

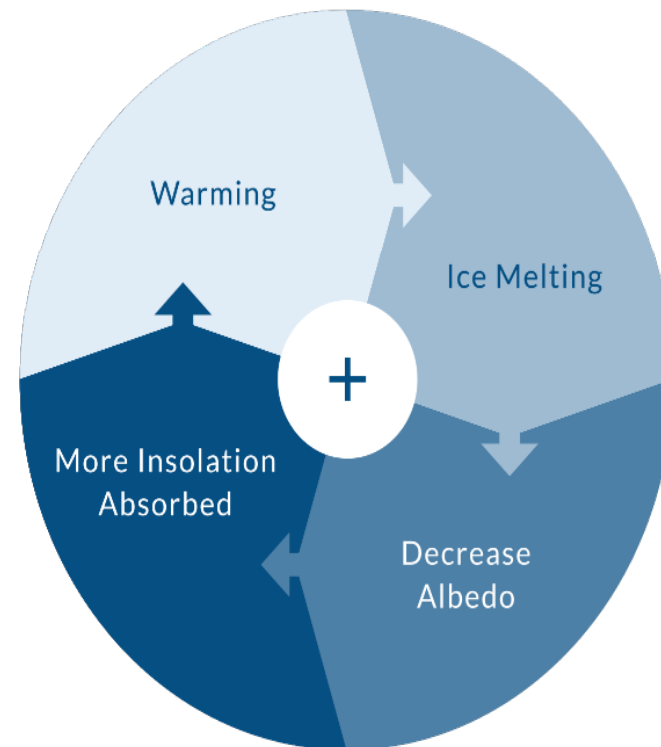
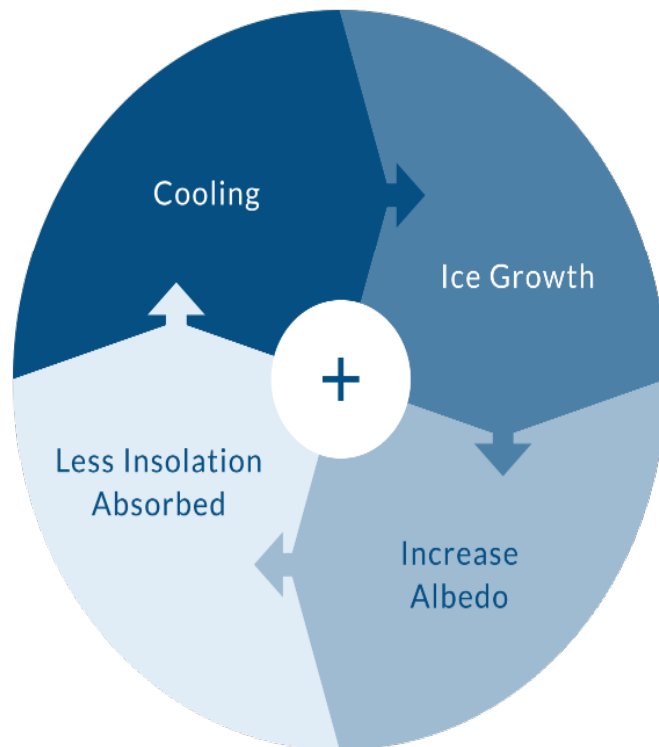


Negative Feedback Mechanism



POSITIVE FEEDBACK MECHANISM

Changes in albedo can create a positive feedback that reinforces a change in the climate. A positive feedback is a process which amplifies the effect of an initial change.



WEATHERING

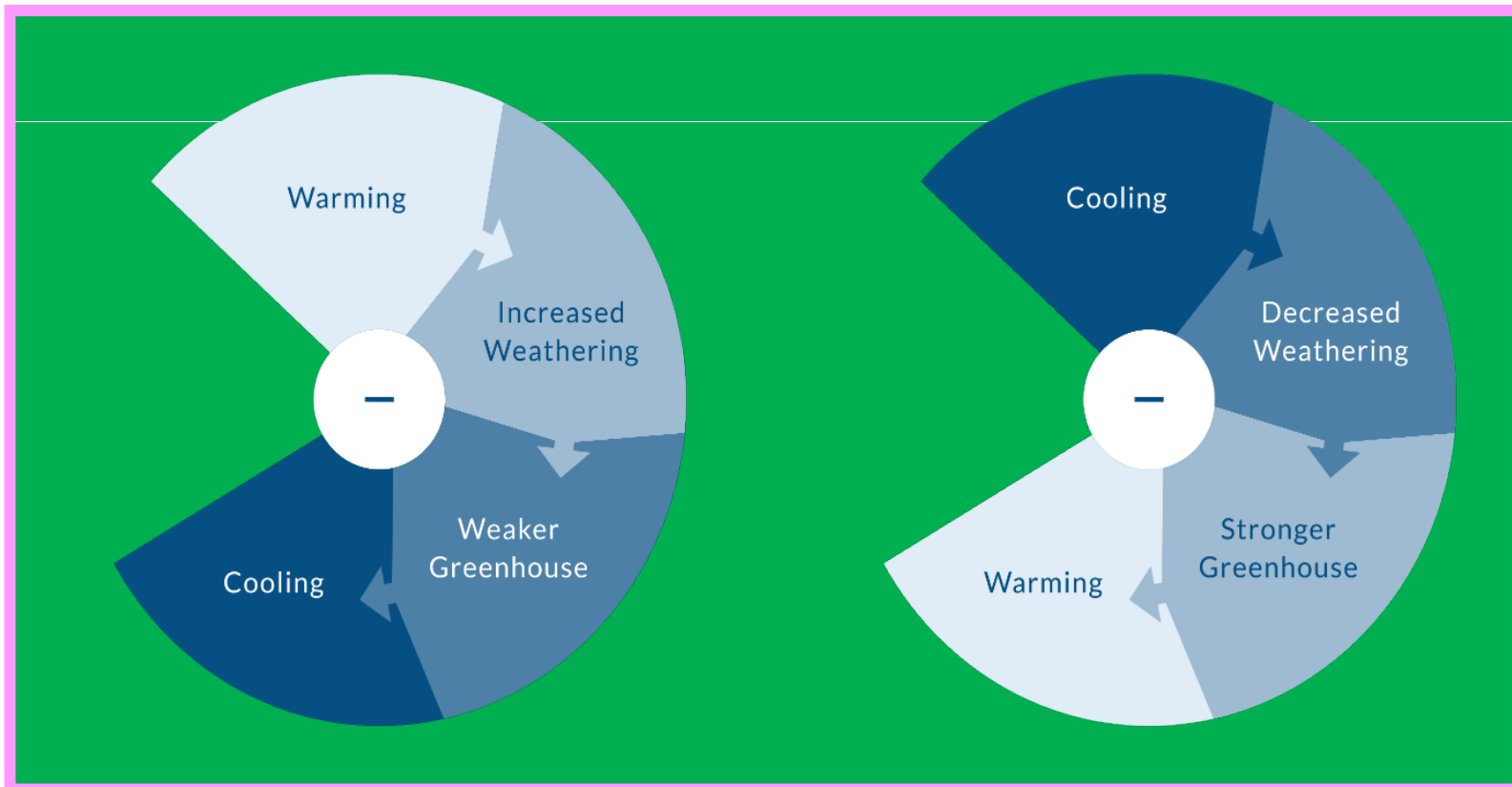
Rocks exposed at earth's surface interact with water and the atmosphere undergoing a set of chemical and physical changes called weathering.

WEATHERING FEEDBACK

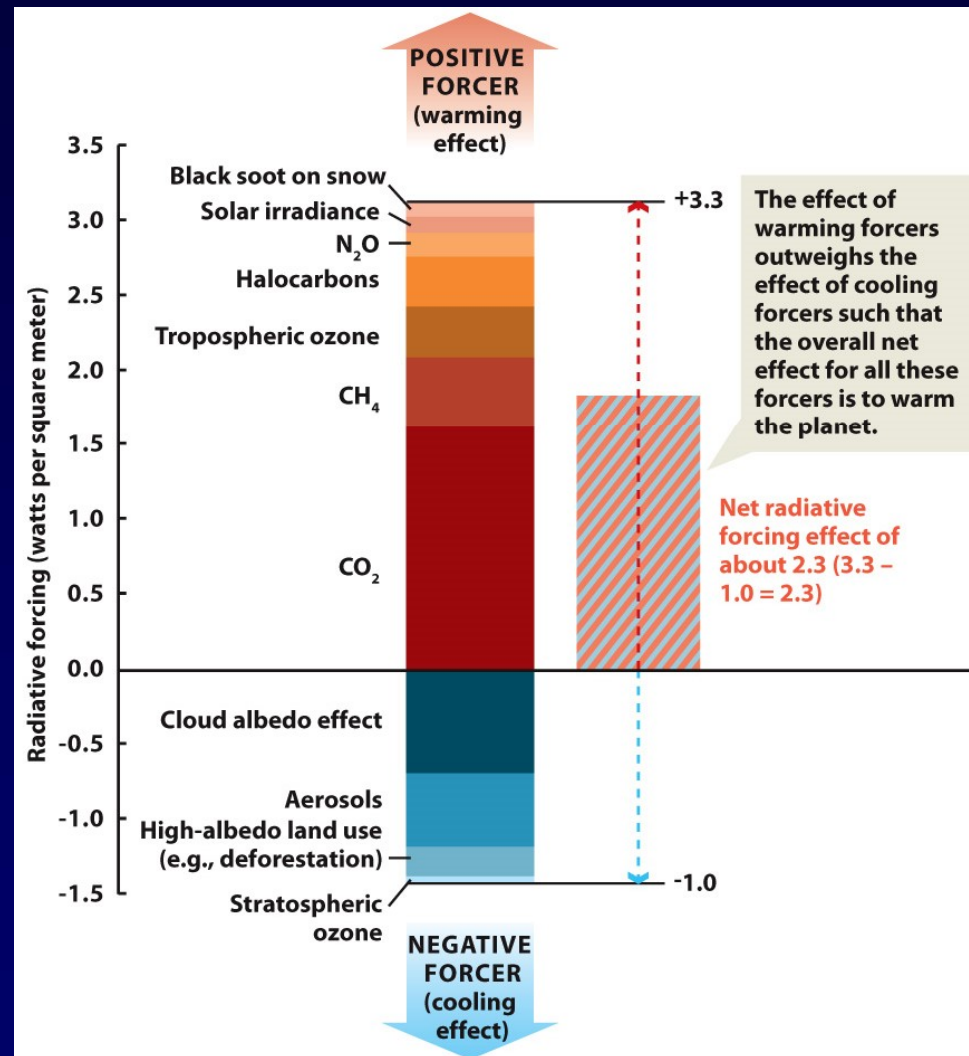
- Weathering process involves a chemical reaction
 - Carbonic acid is formed when carbon dioxide dissolves in water
 - Acid rain contains carbonic acid
 - Another chemical reaction is initiated when acid rain falls on rocks, dissolving minerals and breaking down their structure
- Weathering process is a sink for atmospheric carbon dioxide (an important greenhouse gas)
 - Removing carbon dioxide from the atmosphere weakens the greenhouse effect, cooling the Earth
 - Chemical weathering occurs more rapidly in hotter climates, which are associated with higher levels of carbon dioxide

NEGATIVE FEEDBACK MECHANISM

- In the Earth system, positive and negative feedbacks are essential components that play an important role in maintaining a more or less stable state.
- A negative feedback mechanism stabilizes a system, preventing an extreme state.



Currently the overall effect of climate forcers is warming the planet.



Infographic 21.5

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The Greenhouse Effect

What is it?

The Greenhouse Effect



Some energy is reflected back out to space

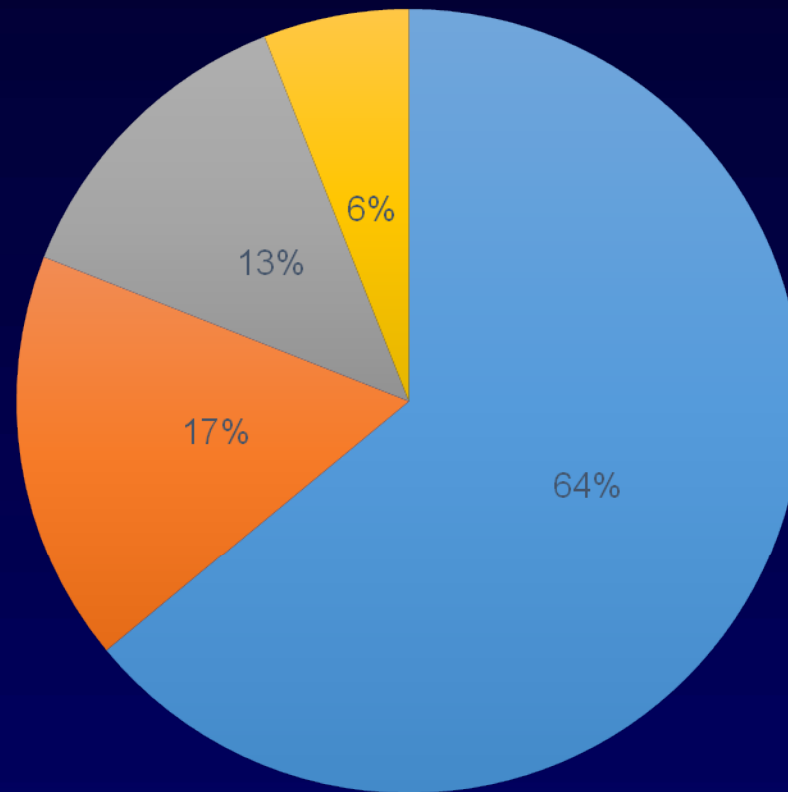
Earth's surface is heated by the sun and radiates the heat back out towards space

Solar energy from the sun passes through the atmosphere

Greenhouse gases in the atmosphere trap some of the heat



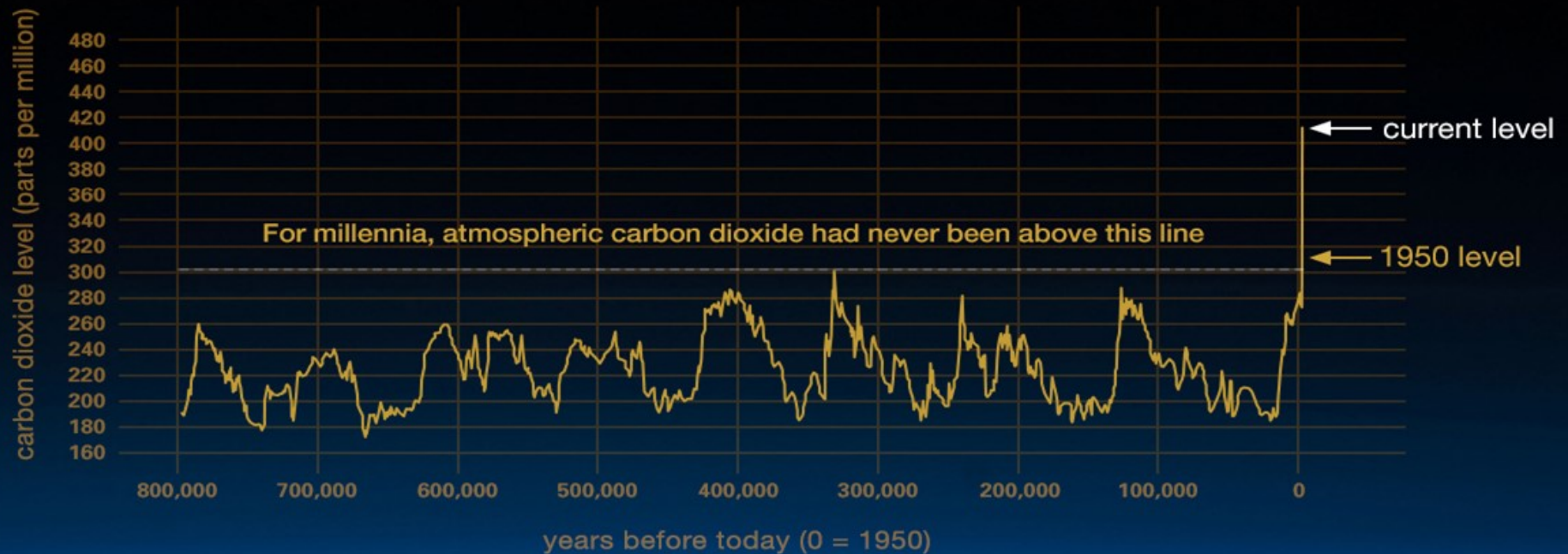
Greenhouse Gases



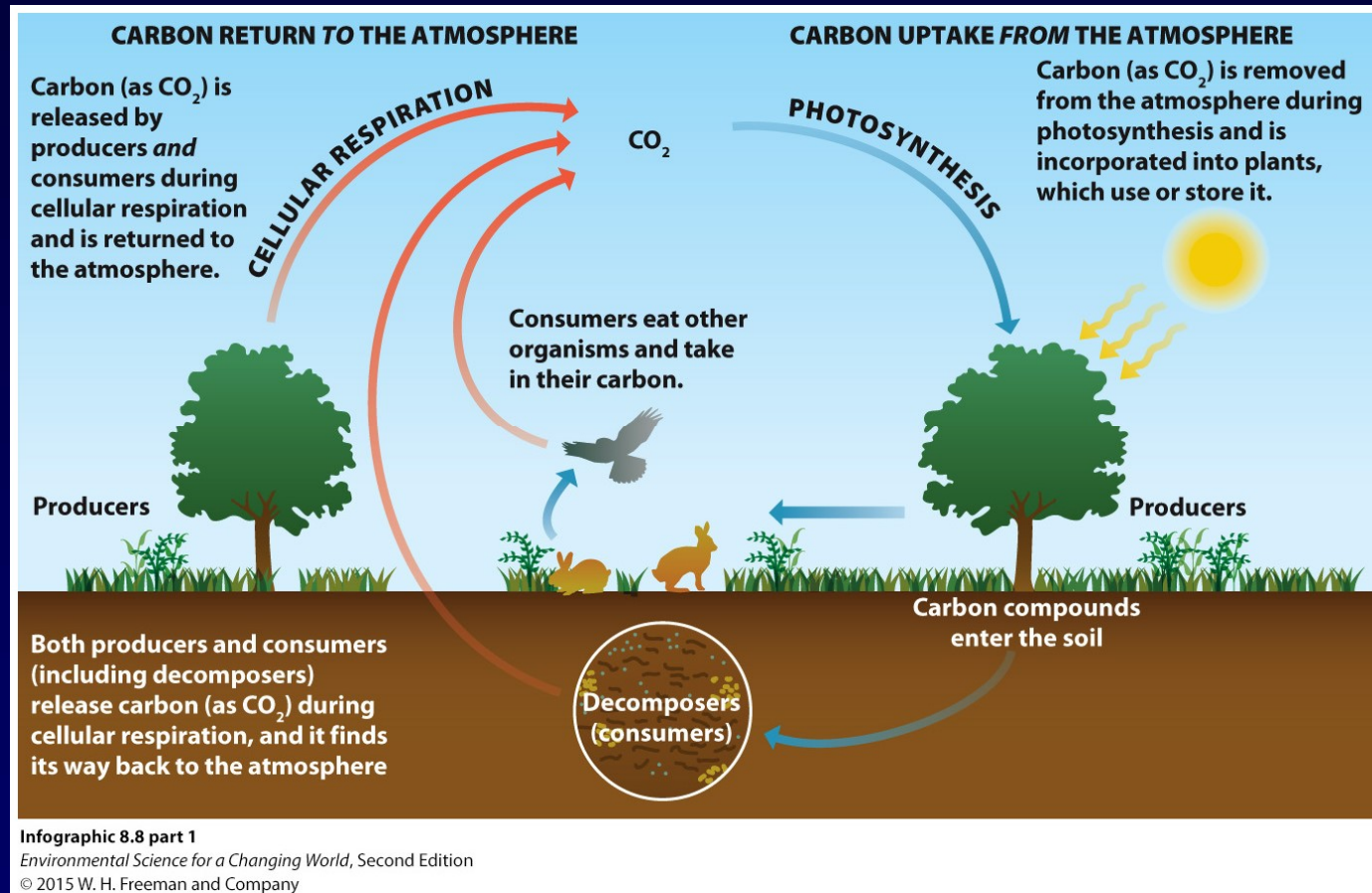
■ Carbon Dioxide ■ Methane ■ Chloroflouro Carbones ■ Nitrous Oxide

- CO₂ concentration in the atmosphere is currently 40% higher than it was when industrialisation began.
- Other greenhouse gases are emitted in smaller quantities, but they trap heat far more effectively than CO₂, and in some cases are thousands of times stronger.

Greenhouse Gases - Carbon Dioxide



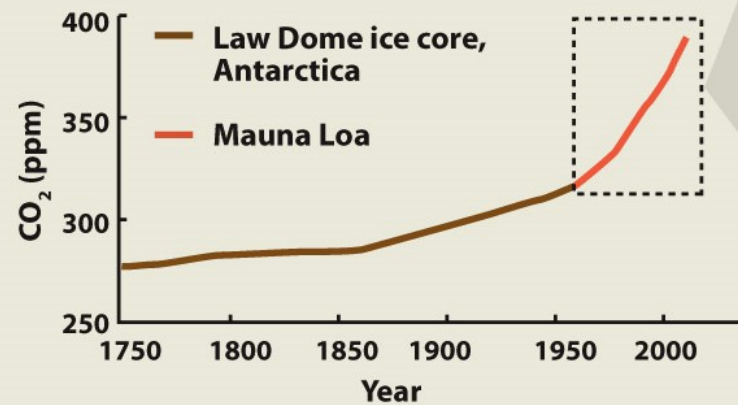
CARBON DIOXIDE IS THE DOMINANT GREENHOUSE GAS



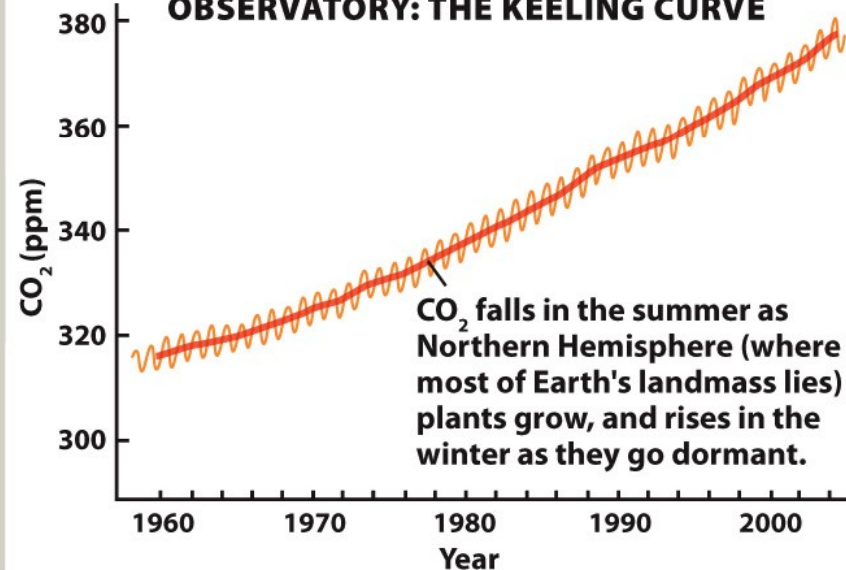
Concentrations of carbon dioxide in the atmosphere are generally maintained by the carbon cycle.

CONCENTRATIONS OF CARBON DIOXIDE HAVE INCREASED FROM PRE-INDUSTRIAL LEVELS OF 280 ppm to ~400 ppm AS OF 2015

CO₂ LEVELS: ICE CORE AND KEELING CURVE COMPOSITE DATA



ATMOSPHERIC CO₂ LEVELS, MAUNA LOA OBSERVATORY: THE KEELING CURVE

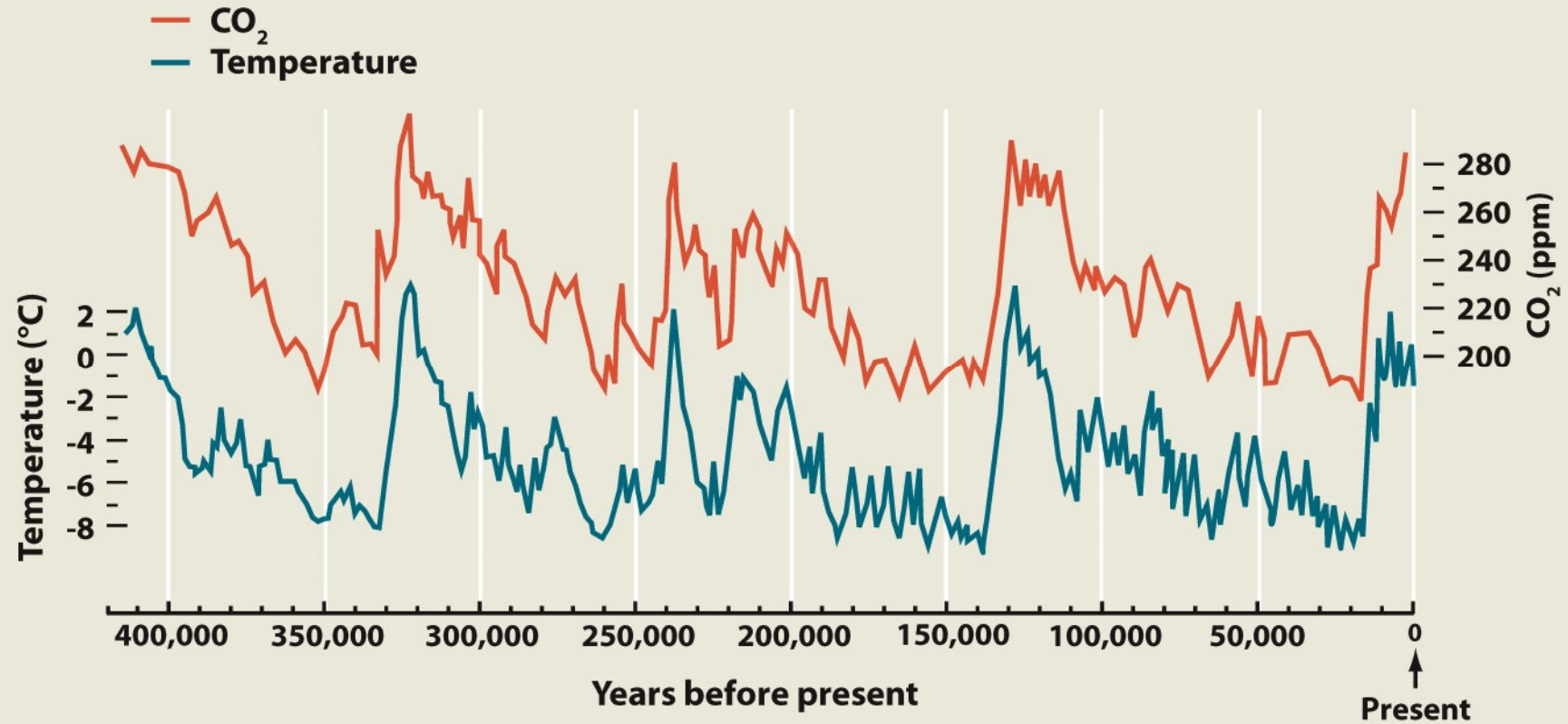


Infographic 21.7 part 1

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TEMPERATURE AND ATMOSPHERIC CO₂ CONCENTRATION OVER THE PAST 400,000 YEARS (VOSTOK ICE CORE)

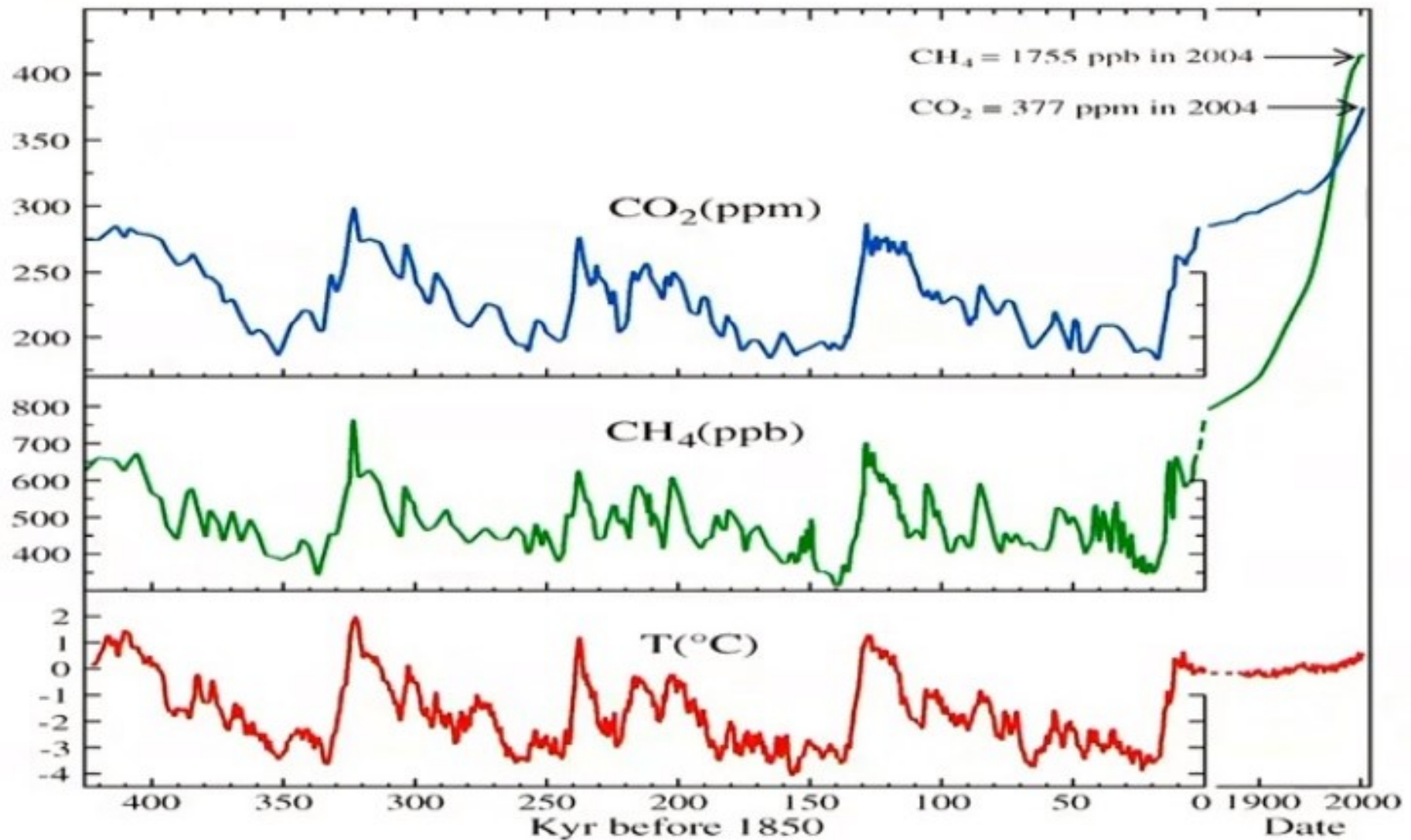


Infographic 21.7 part 2

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Variations in Temperature, Methane, and Carbon Dioxide with Respect to Time.



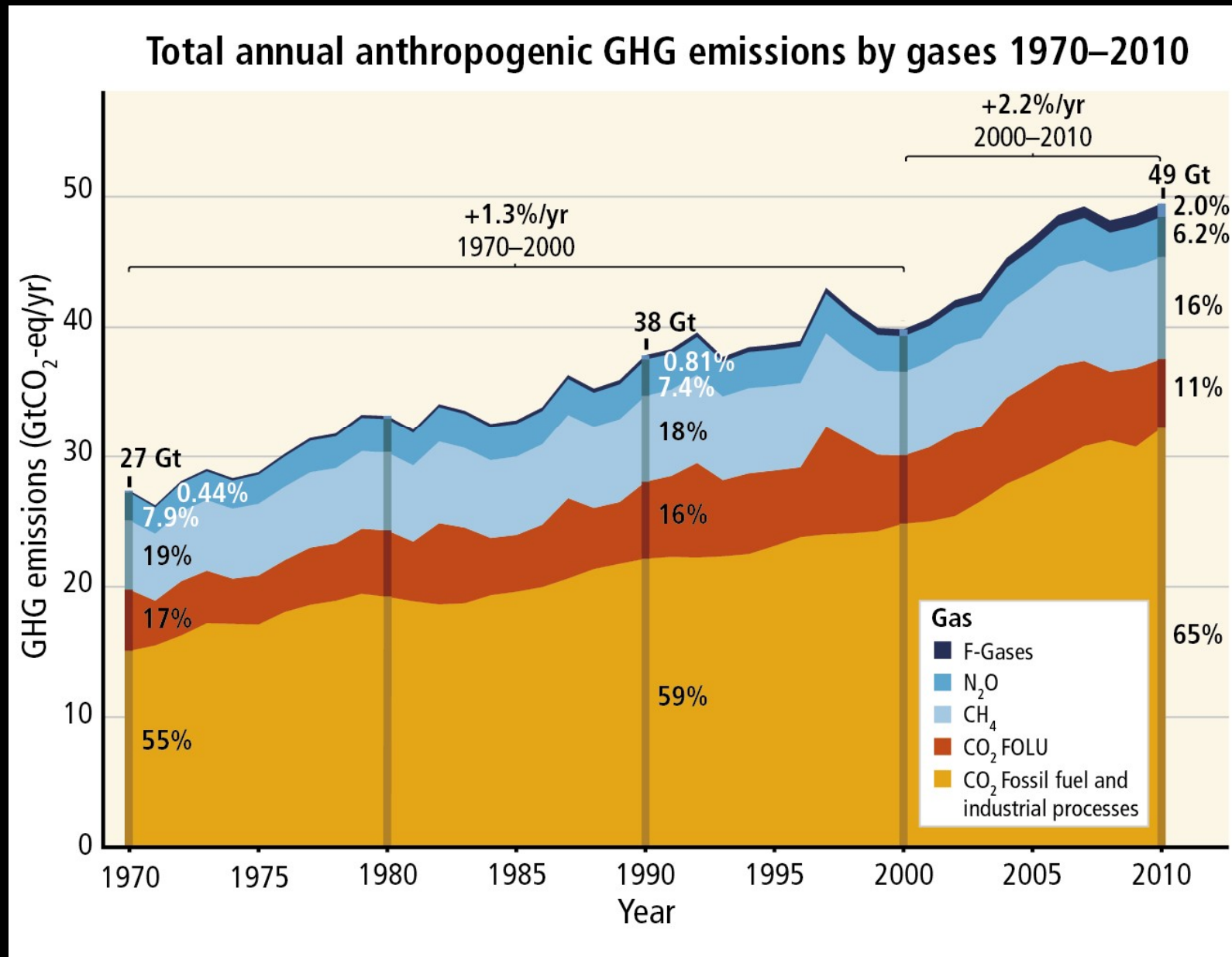
➤ The figure displays variations in methane (parts per billion in volume), carbon dioxide (in part per thousand in volume) and temperature with respect to time (from 450,000 years to present) in a nonlinear scale. Further, the temperature has changed and then carbon dioxide and methane have responded during the geological past i.e. between ~450,000 years and 1840 years (initiation of industrialization) and vice-versa. Whereas, during post industrialization, methane and carbon dioxide have already changed, but temperature has not yet responded as much. This clearly indicates that the process, by which temperature was changing and methane and carbon dioxide were responding and vice-versa, were different in the past than that in the present.

➤ In the past, the amount of varied thermal energy received by the earth from the Sun and also impacted by the earth's orbital changes, plate movement, volcanism etc reflected in the change in temperature of the earth first and then subsequently methane and carbon dioxide changed. But currently, both methane and carbon dioxide increased all time high without causing similar changes in temperature. The processes, giving rise the all-time high concentrations of methane and carbon dioxide, are different. Fossil fuel usage, industrial practices, agricultural practices, change of land cover etc have played important roles in increased concentrations of greenhouse gasses. It appears that other processes (chemical weathering, increased cloud formation, high rainfall, increased aerosols etc) are working as negative feedback to the earth climate system to keep the temperature optimum. Further, the increase in temperature is regulated by the oceans due to high specific heat capacity of ocean water.

The cooling trend is slower than warming trend because of following reasons":

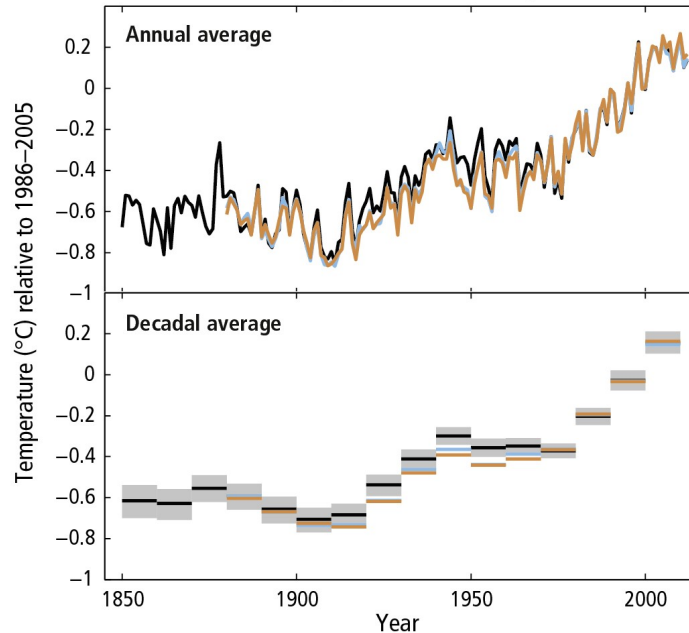
Cooling means reduction in air temperature. This forces humidity of air to reduce and thereby decrease in snow fall and therefore build-up of glacier takes more time. These signify longer elapsed time during the cooling. Whereas, due to increase in insolation, glaciers melt which further resulted decrease in albedo of earth and thereby trapping more insolation and therefore rapid increase in temperature.

What does this graph mean?

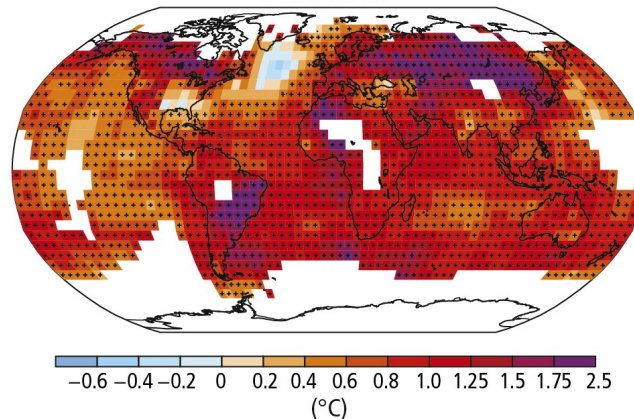


Climate Changes Currently Happening

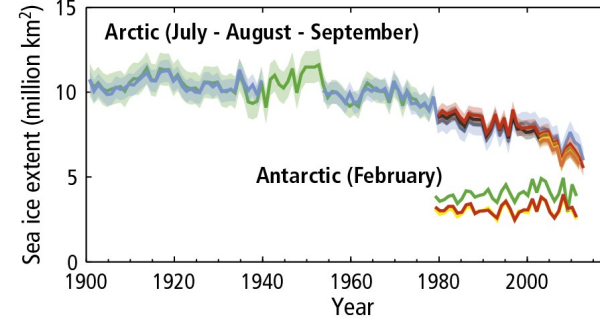
(a) Observed globally averaged combined land and ocean surface temperature anomaly 1850–2012



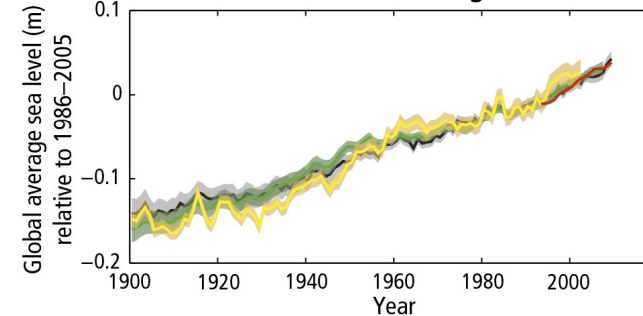
(b) Observed change in surface temperature 1901–2012



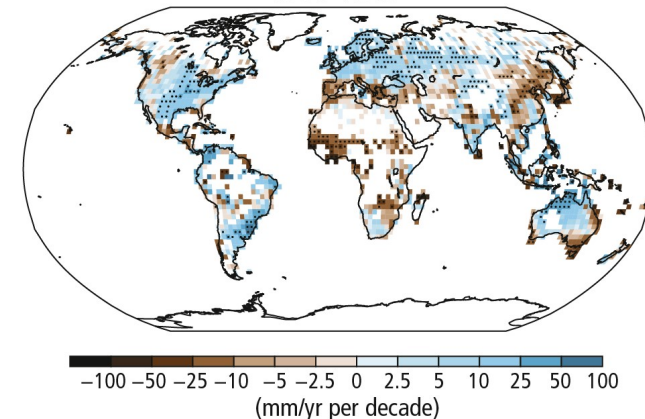
(c) Sea ice extent



(d) Global mean sea level change 1900–2010



(e) Observed change in annual precipitation over land 1951–2010



CONCENTRATIONS OF OTHER GREENHOUSE GASES ARE ALSO INCREASING

GREENHOUSE GASES: RELATIVE CONTRIBUTIONS TO GLOBAL WARMING



Infographic 21.3 part 2

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Different greenhouse gases have different heat trapping abilities.

For example, one molecule of CH₄ traps 25 times the heat of one molecule of carbon dioxide.

Summary on Climate forcing

- Climate change is driven by perturbations to the energy balance of the Earth system. These perturbations are called “climate forcings”. The climate forcings is associated with gases, aerosols, land use, and solar variability.
- Factors that affect climate change are usefully separated into forcings and feedbacks. A climate forcing is an energy imbalance imposed on the climate system either externally or by human activities. Examples include changes in solar energy output, volcanic emissions, deliberate land modification, or anthropogenic emissions of greenhouse gases, aerosols, and their precursors.
- A climate feedback is an internal climate process that amplifies or dampens the climate response to an initial forcing. An example is the increase in atmospheric water vapor that is triggered by an initial warming due to rising carbon dioxide (CO₂) concentrations, which then acts to amplify the warming through the greenhouse properties of water vapor.

Summary on Climate forcing.....Contd.

- Climate forcings can be classified as radiative (direct or indirect) or nonradiative. Direct radiative forcings affect the radiative budget of the Earth directly; for example, added CO₂ absorbs and emits infrared (IR) radiation. Indirect radiative forcings create a radiative imbalance by first altering climate system components, which then almost immediately lead to changes in radiative fluxes; an example is the effect of aerosols on the precipitation efficiency of clouds.
- Nonradiative forcings create an energy imbalance that does not involve radiation directly; an example is the increasing evapotranspiration flux resulting from agricultural irrigation.
- Climate response is the change in the climate system resulting from a climate forcing.

Summary on Climate forcing.....Contd.

- The earth is a planet in dynamic equilibrium, in that it continually absorbs and emits electromagnetic radiation.
- It receives ultra-violet and visible radiation from the sun, it emits infra-red radiation and energy balance says that 'energy in' must equal 'energy out' for the temperature of the planet to be constant.
- This equality can be used to determine what the average temperature of the planet should be.
$$\lambda_{\max} T = \sigma_w$$
- Both the sun and the earth are black-body emitters of electromagnetic Radiation because they are masses capable of emitting and absorbing all frequencies (or wavelengths) of electromagnetic radiation uniformly.
- The distribution curve of emitted energy per unit time per unit area versus wavelength for a black body was worked out by Planck in the first part of the twentieth century.
- First, the total energy emitted per unit time integrated over all wavelengths is proportional to $(T)^4$.
- Second, the wavelength of the maximum in the emission distribution curve varies inversely with T i.e., $\lambda_{\max} \propto 1/T$ or, $\lambda_{\max} T = \sigma_w$ (σ_w is the Wien's const).

Summary on Climate forcing.....Contd.

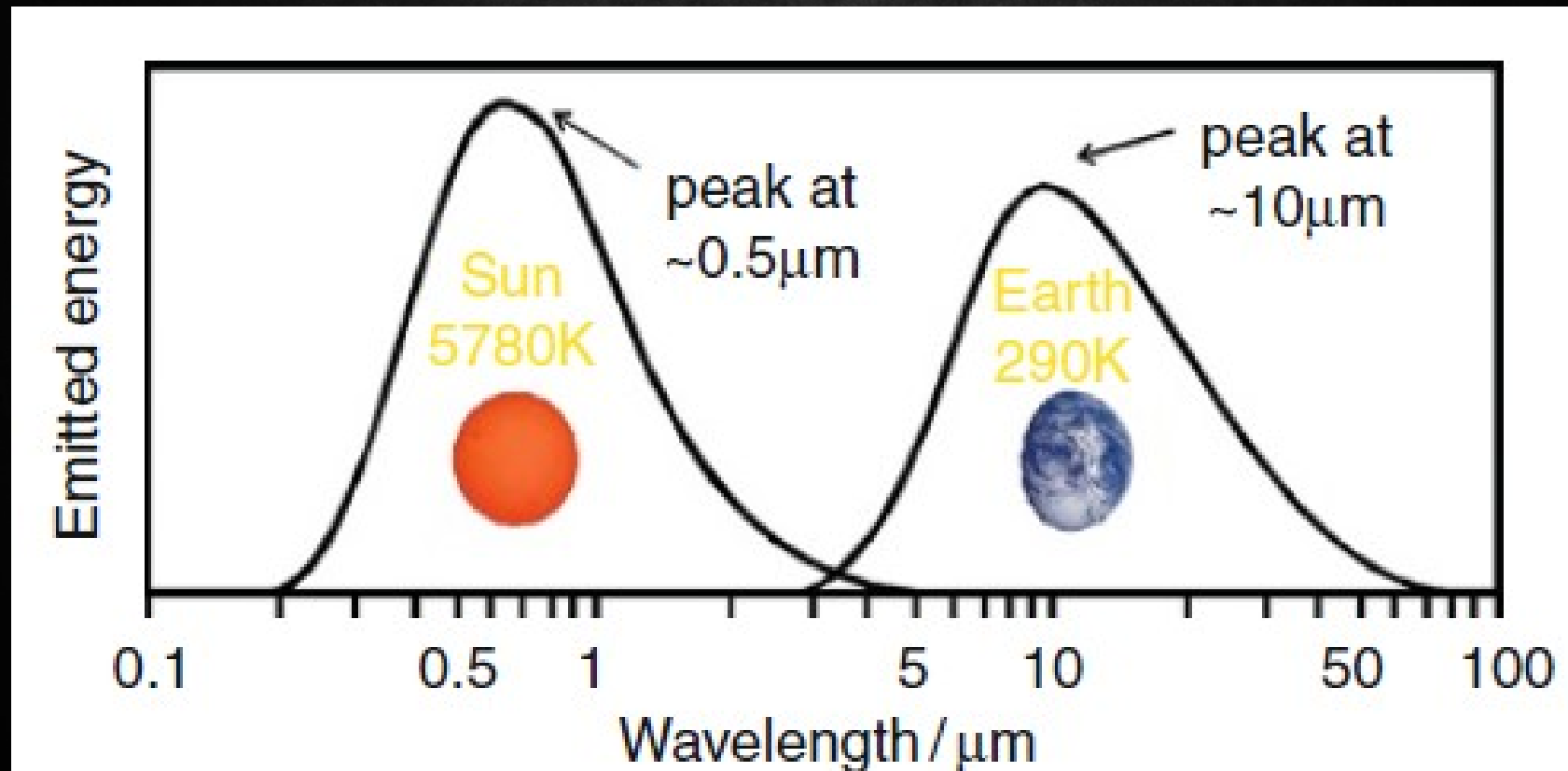
- These are Stefan's and Wien's Laws, respectively.
- The solar flux energy intercepted per second by the earth's surface from the sun's emission can be written as $F_s(1-A)\pi R_e^2$, where F_s is the solar flux constant outside the Earth's atmosphere ($1368 \text{ Js}^{-1}\text{m}^{-2}$), R_e is the radius of the Earth ($6.38 \times 10^6 \text{ m}$), and A is the earth's albedo, corresponding to the reduction of incoming solar flux by absorption and scattering of radiation by aerosol particles (average value 0.28).
- The infra-red energy emitted per second from the earth's surface is $4\pi R_e^2\sigma T_e^4$, where σ is Stefan-Boltzmann constant ($5.67 \times 10^{-8} \text{ Js}^{-1}\text{m}^{-2}\text{K}^{-4}$) and $4\pi R_e^2$ is the surface area of the earth. At equilibrium, the temperature of the earth, T_e , can be written as:

$$T_e = \left[\frac{F_s(1-A)}{4\sigma} \right]^{1/4}$$

Summary on Climate forcing.....Contd.

- Using the data above yields a value for T_e of 256 K. Mercifully, the average temperature of the earth is not a Siberian 17°C , otherwise life would be a very unpleasant experience for the majority of humans on this planet.
- The reason why our planet has a hospitable higher average value of 290 K is the greenhouse effect. For thousands of years, absorption of some of the emitted infra-red radiation by molecules in the earth's atmosphere (mostly CO_2 , O_3 and H_2O) has trapped this radiation from escaping out of the earth's atmosphere (just as a garden greenhouse operates).
- Some is re-radiated back towards the earth's surface, thereby causing an elevation in the temperature of the surface of the earth. Thus, it is the greenhouse effect that has maintained our planet at this average temperature, and for this fact we should all be very grateful!
- This phenomenon is often called the 'primary' greenhouse effect. It is, therefore, a myth to portray all aspects of the greenhouse effect as bad news; it is the reverse that is true.

Black-body emission curves from the sun ($T \sim 5780 \text{ K}$) and the Earth ($T \sim 290 \text{ K}$), showing the operation of Wien's Law that $\lambda_{\text{max}} \propto (1/T)$.



The two graphs are not to scale.



Pillow lava at CIR observed during manned submersible dive